Market Response to Rate Decisions: Forward Guidance and Cross-Asset Reactions to FOMC Announcements, 2008–2026

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Abstract

The Federal Reserve is arguably the pacesetter for all central banks around the world due to its influence over the most dominant economy in the world. One of several tools in the Fed's toolbox is the policy to determine the federal fund rate, a rate fundamental in dictating the affordability of credit and short-term loans. How are the capital markets affected when the Fed announces to cut, hike, or hold rates steady for the sake of the US economy? In this paper, utilizing the constant-mean model, statistical findings show that equity markets already price-in the expected policy rate of each announcement, leaving only the surprise component to drive a volatile reaction in all markets. Further, treasuries are the only instruments that experience sharp declines when policy rates are held steady, indicating nuance in market expectations and efficiency, fixed-income convexity, and investor outlook for the future.

I. INTRODUCTION

The American economy in the 21st century has undergone various economic cycles that cascaded into worldwide events, from the 2008 sub-prime mortgage crisis to the COVID-19 pandemic, two precursors that prompted unexpected monetary policy from the Federal Reserve, whose sole responsibility is to bring the stability to the economy and banking system. This intended role to anticipate and drive out recessions and other instabilities can indirectly cause distress to the asset markets. The Fed has various tools at its disposal to utilize. One of the versatile tools at its disposal is the influence over the federal fund rate (also known as the policy rate) which is decided by the Federal Open Market Committee (FOMC). The FOMC is a composed of 12 voting members across the United States that plan and calculate monetary policy decisions, months to years in advance. They meet typically eight times a year with majority vote deciding the policy rate. Every scheduled decision is announced at 2PM ET which is relevant for the structure of this paper and introduces the 3-day window, where we will analyze the anticipation reaction

of the asset markets, three days before the announcement, intraday announcement reaction, and post-announcement reaction, also known in the literature as monetary momentum noted by Neuhierl and Weber (2018). On rare occasions, there can be unscheduled FOMC meetings, synonymously referred to as emergency meetings, set to tackle catastrophic economic conditions such as the lock-downs caused by the COVID-19 pandemic. These scenarios provide systematic risk collectively to virtually all markets, each with different magnitudes of volatility. Hence, the objective of this paper, using the constant-mean model, is to statistically test and measure the performance, reaction, and change of each respective instrument in different markets; all encapsulated in an event study framework to observe cross-asset sensitivity changes with respect to every FOMC announcement across the sample. The asset classes of focus will be major equity indices (S&P 500, Russell 2000, and a Financials index), treasuries (2Y to 30Y), precious metals (gold and silver), the US dollar, and the CBOE Volatility Index ($\hat{\text{V}}$IX). The timeline of the study will begin at the financial crisis, January 2008 until April 2026, providing multiple unique Fed regimes throughout, like zero-interest rate environments. An overall moderate sized study across various correlated and uncorrelated assets will provide not only interesting results but statistical findings that buttresses much of the concurrent finance theory.

The inspiration for this paper was heavily influenced by the work of Bernanke and Kuttner (2004) where they discovered how the stock market is affected by federal reserve policy, more so, utilizing a regression model as the fundamental framework: $i_e = \alpha + \beta \cdot (surprise_t) + \epsilon_i$, where they found that an unanticipated 25-basis-point rate cut raised the S&P 500 by approximately 1%. The findings of Bernanke and Kuttner were quantified in a standard framework, the marginal unit increase in asset prices when an unexpected rate reduction occurred. Tangential to their work, the findings in this paper will be computed using a mean-comparison event study framework, where we will look to answer if the selected set of assets react in a meaningful way to policy rate cuts, hikes, and holds on average using t-tests. With that said, the major three hypotheses are as follows: Does the FOMC decision type (hike, cut, or hold) make a detectable difference to how each market reacts, does each respective asset actually respond in a significant way to a certain decision type in a particular regime, and lastly, is there a difference in how assets react to scheduled meetings in contrast to unscheduled or emergency meetings. In the findings, we find that the fixed-income market, specifically, the 2Y, 5Y, and 10Y treasuries declined significantly for FOMC hold meetings, major equity indices did not show any significant reaction in any decision type in the pooled sample, and unscheduled emergency meetings coincided with extreme market movements

(such as −5.3% S&P 500, +7.6 VIX), where scheduled meetings did not, on average. Next, we will discuss the methodology used to collect and process data, the software used, and the construction of the study.

II. METHODOLOGY

The primary methodology of this paper is an event study framework inspired by MacKinlay (1997) over the time horizon of January 2008 to April 2026, where 150 FOMC announcements have been recorded in the total sample. The market reaction and price sensitivity will be analyzed on a standard-discrete time window function, [-1,1], encapsulating one day before, intraday (announcement day), and one day after the announcement. For robustness, the event window will stretch up to 5 days after, [0,5], a particular announcement to see how much markets move post-announcement.

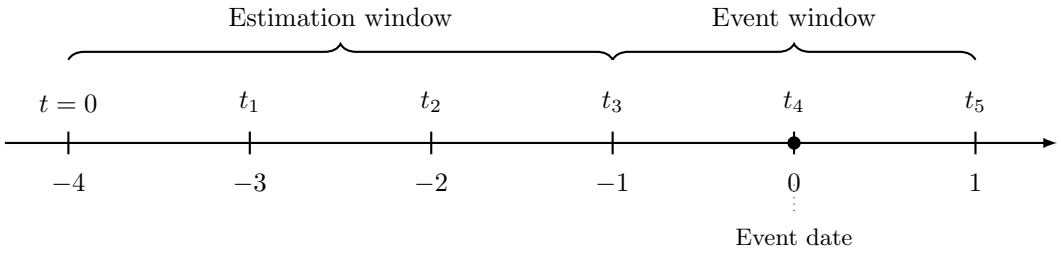


FIG. 1. Event study time window.

To measure security performance effectively, there will be different metrics used for every asset of interest. First, for major equity indices, such as the S&P 500, the calculated measure will be cumulative abnormal returns (CARs) which is the most relevant metric for equities, as it best captures the change in actual returns with respect to expected returns of the investor, therefore, CAR will be used to quantify nominal equity price changes. The time window for equity indices will expand to 120 days before an announcement and 20 days after, [-120,20], to observe possible trends in the CAR, pointing to market anomalies.

Cumulative average return formula as stated:

$$CAR = \sum_{t=t_1}^{t^2} AR_{i,t} \tag{1}$$

Where, AR is the abnormal returns, R_i is the return of the stock, and $E(R_i)$ is the expected return of

the investor.

$$AR = R_i - E(R_i) \tag{2}$$

To derive the expected return of the stock of interest, the standardized capital asset pricing model will be used. Further, R_f is the risk-free rate, β_i is the volatility measure of the stock, and $E(R_m)$ is the expected return of the market.

$$E(R_i) = R_f + \beta_i(E(R_m) - R_f) \tag{3}$$

When analyzing our next form of security types; treasuries, VIX, commodities, and foreign-exchange series, we will simply calculate the cumulative raw change across each event window. This is due to the fact that these securities do not experience the same systematic drift that major stock indices experience (where there is inter-correlation risk among separate global and domestic indices such as the NASDAQ and S&P500). Moving onto the FOMC announcements, the standard announcements are either one of three options, to hike, cut, or hold the federal fund rate by a variable amount of basis points. On rare occasions, these announcements can be unexpected and emergency-based decisions enacted to deal with a dire economic conditions. This introduces the cautionary problem of endogeneity caused by omitted variable bias which limits the inference to be made on any statistical findings, for example, if the Federal Reserve raises policy rates drastically and the equity markets shortly after diminish, we cannot make a direct casual inference and say the policy rate increase caused the stock market to fall. Further discussion on endogeneity will be provided later on. Lastly, the main events of interest will be seven major events within the sample: the Global Financial Crisis (GFC), Zero Interest Rate Policy (ZIRP), Normalization, Mid-Cycle, COVID, Tightening, Plateau, and Easing. These will be classified as the major Fed regimes, setting the groundwork for how different asset classes react to different interest rate environments. For statistical analysis, we will use Welch's t-test to see if there are statistically significant changes for how markets react to contrasting monetary policy (e.g. rate vs hike) and to test scheduled vs unscheduled mean reactions for each asset. The one sample t-test will also be used to test whether there was a significant and measurable effect from a certain FOMC decision or not. For additional robustness testing, there will include bootstrapping confidence intervals, with 10,000 resamples, all with the purpose of flattening outliers or skewed data that may strife the final results.

A. Data Collection & Setup

In preparation for the event study, all relevant data was collected and aggregated from public repositories; for monetary data, the Federal Reserve Bank of St. Louis (FRED) provided time series data including treasury yields and federal fund rates. For Market price data, extraction from Yahoo Finance was adequate for daily security prices. All data was pulled from both repositories using Python 3.12 with the assistance of libraries (pandas and NumPy), downloaded and cached as CSV files for data manipulation and hypothesis testing. At this point, the construction of our event study can be broken down into five concise phases, all programmed within Python. Firstly, a daily panel is built by merging all FRED and Yahoo Finance time series data into a business-day index (excluding holidays and weekends). In addition, log returns and yield changes were also computed at this stage. Secondly, a FOMC calendar is constructed from a hand-verified list of announcement dates and cross-validated against FRED target-rate changes. It was vital to verify that every observed interest rate change in the sample window corresponded to a meeting in the calendar within a one-business-day tolerance (because new targets are always recorded on the business day following the initial 2:00 PM ET announcement). Third, after each meeting is announced, there must be a computation of cumulative returns and yield changes over each of the four event windows ([0,0] (same-day), [-1,+1] (3-day standard), [0,+1](2-day standard), [0,+5] (6-day standard)), and for the equity series, cumulative abnormal returns is computed utilizing the standard constant-mean model. Fourth, hypothesis tests are conducted at the standard p-value of $\alpha = 0.05$. Welch's two-sided t-tests will be assumed and results will be organized in order of significance from 1%, 5%, and 10% levels. Finally, as previously mentioned, there will include bootstrapped 95% confidence intervals using 10,000 percentile-method resamples (using a fixed random seed per each resample). Bootstrapping allows us to test the robustness of the hypothesis with the limited amount of real data. Lastly, in the fifth phase is data visualization, where figures and tables are generated from the outputs, please note that all figures will be watermarked due to authenticity reasons as an independent researcher.

B. Asset Types, Regimes, and Categorization

Cross-validation was treated as a first-class step in the pipeline, therefore, during development; the calendar-versus-FRED cross-check surfaced a subtle timing bug in which meeting-day rate look-ups were reading pre-announcement targets rather than post-announcement targets, which would have miscoded

roughly one-fifth of all meetings had it not been caught before downstream analysis. All code and work, along with the intermediate CSV outputs and the reproducible figures, is stored and organized in a version-controlled repository in standard files. Within the empirical data, there are four classes of assets listed in the table below. In the sample of n=150, all FOMC announcements between January 22, 2008 and April 29, 2026 are accounted, including 146 scheduled meetings and 4 unscheduled/emergency actions (January 22, 2008; October 8, 2008; March 3, 2020; and March 15, 2020).

TABLE I. Assets of Study

Asset	Symbol	Asset Class
S&P 500	GSPC	Equity
Russell 2000	RUT	Equity
Financials	XLF	Equity
Gold	GC=F	Precious Metals
Silver	SI=F	Precious Metals
Dollar	DXY	Currency
2Y yield	DGS2	Treasuries
5Y yield	DGS5	Treasuries
10Y yield	DGS10	Treasuries
30Y yield	DGS30	Treasuries

For equities, there are three indices to look at, the well-known S&P 500 index ($\hat{\text{G}}$SPC), the broad Russell 2000 index ($\hat{\text{R}}$UT), and a Financial index, more specifically, the State Street Financial Select Sector, SPDR ETF ($\hat{\text{X}}$LF), which is an exclusively rate-sensitive sector proxy (inversely correlated to interest rates). These indices give our study exposure in all relevant sectors of the stock market. For fixed income securities, we will analyze four distinctive treasury yields: treasury yields at the 2-year (DGS2), 5-year (DGS5), 10-year (DGS10), and 30-year maturity (DGS30). To see the change in the US dollar, the ICE U.S. Dollar Index (DX-Y.NYB) will be utilized. Commodity prices are included in the empirical data but with only two major precious metals, front-month gold (GC=F) and silver (SI=F) futures. Lastly, a measure of market volatility can be effectively captured with the CBOE volatility index ($\hat{\text{V}}$IX). Beyond market data, the second frontier of data is derived from the Federal Reserve data bank, mainly federal fund rate historical data. All series of data are observed at daily frequencies over January

1, 2008, through the most recent available date, April 29, 2026. It should be noted that the M1 money supply was considered as variable in the event study but incompatible due to only monthly records of the data being found, rather than daily. All and all, this concludes the data procurement which will we will examine with rigorous statistical methods.

In terms of hypothesis testing, we want to observe real changes in cumulative changes and CAR with respect to the Fed decisions, and for scheduled or unscheduled testing. The optimal statistical method will be to compute Welch's t-statistics to test if there is a true change in the averages. Alongside, two Welch t-tests, there will be a single one sample t-test in addition.

TABLE II. Table Note for Hypothesis Testing

Asset type	Mean (average)	Units
S&P 500, Russell 2000, Financials	Cumulative abnormal returns (CAR) over the [-1, +1] window	Percent (%)
Gold, Silver, Dollar	Cumulative log returns over the [-1, +1] window	Percent (%)
2Y, 5Y, 10Y, 30Y yields	Cumulative yield change over the [-1, +1] window	Basis Points (bp)
VIX	Cumulative point change over the [-1, +1] window	Index points

The following table will prove useful for navigating context in each particular Fed regime, complementary to the findings in the results section. Also, please note that the term 'federal fund rate' and 'policy rate' will be used interchangeably throughout the paper, as they have the same meaning (the same logic applies to 'emergency meetings' and 'unscheduled meetings').

TABLE III. Monetary Policy Regime Definitions

Regime	**Period**	**Characterization**	***n***
GFC	Jan 2008 – Dec 2008	Global Financial Crisis; emergency cuts; ZIRP onset	10
ZIRP	Jan 2009 – Nov 2015	Zero interest rate policy; QE1/QE2/QE3; forward guidance	55
Normalization	Dec 2015 – Jun 2019	Gradual hiking cycle; balance-sheet runoff begins	29
Mid-cycle	Jul 2019 – Feb 2020	Insurance cuts amid trade tensions	5
COVID	Mar 2020 – Dec 2021	Emergency cuts; return to ZIRP; QE4	16
Tightening	Jan 2022 – Jul 2023	Aggressive hiking cycle (525 bp cumulative)	13
Plateau	Aug 2023 – Aug 2024	Terminal rate held at 5.25–5.50%	8
Easing	Sep 2024 – Present	Cutting cycle begins with 50 bp cut	14

Notes: Regime boundaries are set at the last announcement of each period. n denotes the number of FOMC announcements (scheduled and unscheduled combined) in each regime. Total $n = 150$.

III. RESULTS

We begin by examining the interest rate environment in the sample which spans across the last 18 years. There are seven significant moments to be highlighted; in the first period, there is a violent fall in interest rates clearly from the post-2008 subprime mortgage bubble collapse; federal fund rates fell from 3% in September 2008 to a near-zero range of 0-25% in the following 3 months, leading to December 2008. The second period displays the near-zero range of interest rates that were held for seven years, defining the zero-interest rate response (ZIRP). Although, interest rates began to hike back to normal rates in 2016 to 2019, leading to the third period. In the fourth period, is the inception of the year 2020, where rates once again were cut to near-zero levels, this time for only two years. In the fifth to sixth period, there was unprecedented and historical rate hikes as a hawkish response to inflation, where, in March 2022 rates started to rise (from 0-25%) and peaked in October 2022 at 3.75%. The last period

shows federal fund rates falling from 5.50% to 5.25% to just above 3% in early 2026. In aggregate, it has been a profoundly volatile track record for the Fed which will provide profound results in relation to the markets.

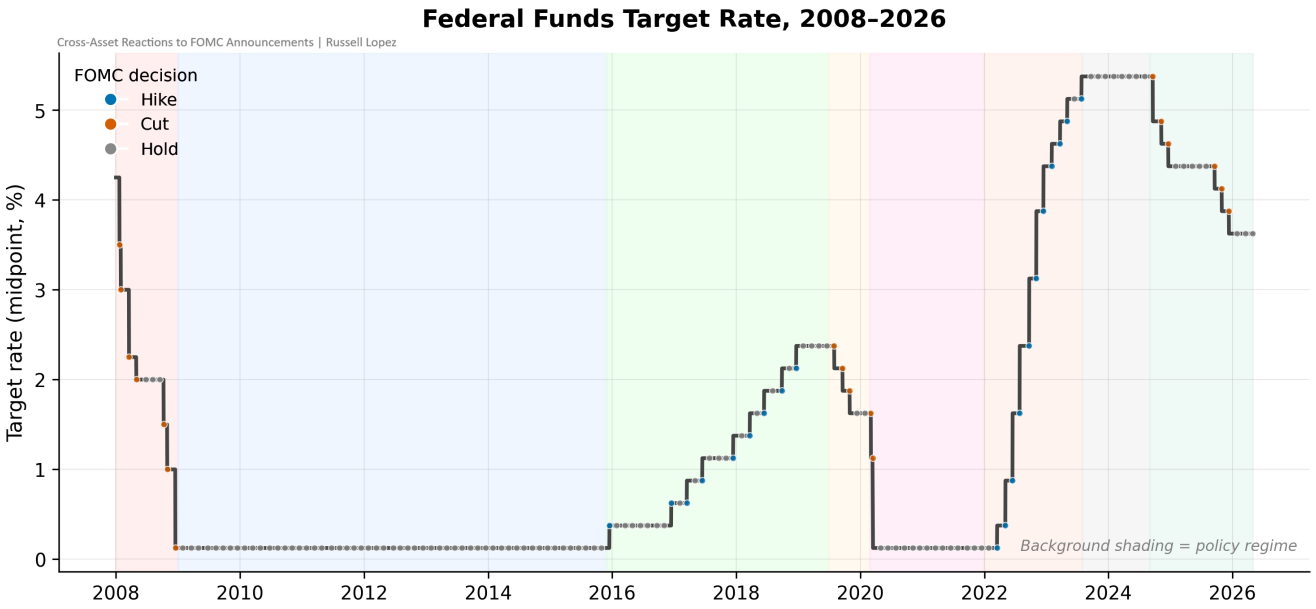


FIG. 2. Federal funds rate timeline

The first hypothesis table (refer to appendix) displays the testing of Welch's two-sample t-test ($\alpha = 0.05$) to test the mean reaction for the two groups ($H_a : \mu_A \neq \mu_B$, where, A and B are different policies), in other words, is there a statistically significant and detectable difference in the averages between two contrasting policies (A vs B) for each respective asset and its metric across different samples. In the first table of the appendix, it is evident that overwhelmingly most mean reactions do not show a statistically significant, detectable difference. This means that regardless of the policy decision enacted for most samples, the asset of study in not impacted. For instance, for the full sample and GFC (global financial crisis) sample, equities show no detectable change in cumulative abnormal returns regardless of if policy rates are hiked versus cut, hiked versus held, or vice versa. The same goes for precious metals and the US Dollar cumulative percentages, treasury yields cumulative basis points, and the VIX index cumulative points (all these assets lack significance at $p > 0.05$). This trend in the data continues for all of the following samples with notable exceptions that will examined momentarily; normalization, mid-cycle, COVID-19, tightening cycle, and easing cycle all for the most part show statistically insignificant changes

to the averages of the S&P 500, Russell 2000, Financial, Gold, Silver, US Dollar, and the VIX index. In terms of notable findings, there are several interesting observations in descending order of statistical significance: first, the few subsequent samples that indicate the strongest evidence of mean differences between their two groups (p¡0.01), is the 5Y (p=0.002) and 10Y (p=0.002) cumulative yield change over the [-1,+1] window; clearly, during the COVID-19 sample, ZIRP was enacted and the 5Y and 10Y treasuries saw a detectable change in yield difference for policy rate cuts versus when rates were held steady. This trend of statistically significance continues for different subsamples, where the 30Y treasury yield in the mid-cycle sample shows to be highly sensitive to rate cuts ($\mu_a = -13.667$) versus holding rates steady ($\mu_b = 2.000$). Similar sensitivity to the averages between two groups also appear in the easing sample, where, the 10Y (p=0.028) and 30Y (p=0.016) have radically large differences in cumulative yield change for rate cuts versus holds (10Y: $\mu_a = 7.50$ and $\mu b = -1.375$, 30Y: $\mu_a = 6.833$ and $\mu_b = -1.750$). All the statistically relevant finds so far, signal real reactions which will be explained in greater detailed in the discussion. Lastly, are the assets with weak reactions ($0.1 > p > 0.05$) which do not warrant enough evidence to make any conclusive claims. Thus, the following display minor changes in their respective averages: in the normalization sample, the US dollar (p=0.074) and 30Y (p=0.096), during the COVID cycle, the 5Y yield (p=0.066), in the tightening cycle, the 2Y yield (p=0.086), and finally, in the easing sample, the 5Y yield (p=0.082).

The findings of the second hypothesis can be found on the regime summary table (table 11 in the appendix) which acts more of a robustness testing. Where, what is being tested is the magnitude of asset reactions with respect to monetary policy decisions, in this test, we are running one sample t-test against a mean of zero ($H_a : \mu \neq 0$) for every possible FOMC decision under every Fed regime. Put simply, will the announcement produce a detectable difference, non-zero reaction (p¡0.05)? If not, then there is no significant market reaction, and the corresponding announcement has no measurable effect (p¡0.10). Naturally, the results are cumbersome, therefore, findings of only true statistical significance will be reported in descending order: in the full sample, the FOMC decision to hold rates steady caused the 2Y (p=0.009), 5Y(p=0.005), and 10Y(p=0.015) treasuries to see large changes in their cumulative yields, as well as excluding zero from their 95% confidence intervals. In the same sample but different decision, when combining the rate cut, hike, and hold policy into an average classified as 'all' decision, and testing it on the full sample, all treasuries by and large react in a statistically significant way; including the 2Y (p=0.006), 5Y (p=0.004), 10Y (p=0.012), and 30Y (p=0.033) cumulative treasury yields. Moving forward, the

additional samples further demonstrate that treasury securities are the most sensitive to FOMC decisions in their respective sub-selective sample testing. In the normalization sample, 30Y (p=0.002) treasury yield have a detectable difference when the Fed decides to hike rates in addition to the 5Y (p=0.0028) treasury yield when rates are held, and the 10Y (p=0.025) and 30Y (p=0.003) treasury yields when the 'all' decision is applied. In the mid-cycle sample, the 30Y (p=0.030) treasury yield is observed to change when rates are cut. Also, for the first time, an equity type is shown, the S&P 500 (p=0.025) sees a non-zero reaction in its CAR when rates are held. The next significant sample is in the tightening period (aggressive 5.25bp federal fund rate), silver surprisingly sees a detectable difference in its cumulative change in price when rates were hiked. Why gold does not see a similar reaction will be theorize in the discussion section. In the second last sample, the plateau period (terminal rates held at 5.25-5.5%), we can see that the treasury yields once again see the only statistically significant changes when rates are held for the 2Y (p=0.021), 5Y (p=0.027), and 10Y (p=0.037). Moreover, when all three FOMC decisions are averaged, the 2Y (p=0.021), 5Y (p=0.027), and 10Y (p=0.037) once more, see a non-zero reaction in their cumulative yields. Finally, for the last sample, the most concurrent being the easing period (cutting cycle after plateau), only the long-term treasuries, 10Y (p=0.048) and 30Y (p=0.036), see a significant reaction in their cumulative yields. These results conclude that treasury securities are the most sensitive to certain FOMC decisions enacted by the Fed in depending on the particular Fed regime.

The last hypothesis is stored in the table below, once again, using Welch's two-sample t-tests to test if the mean reaction of each respective asset is different from a scheduled FOMC meeting versus an unscheduled or emergency FOMC meeting. If statistical significance (p¡0.05) is found, then the difference will be detectable and significant. Now, in the findings, there is no statistically significant p-values, this presents the limits of statistical testing and presents the endogeneity problem. Although, despite lack of findings, there is still interesting results to be observed in the mean differences for each asset when FOMC meetings are scheduled versus when meetings are unscheduled. For example, the equities indices see huge variance in their differences; 5.319% for the S&P 500, 7.81% for the Russell 2000, and 7.7125% for Financials. Silver experiences a larger jump at 7.626%, in contrast, its counterpart, gold is at 3.3227%. The most dramatic changes occur in the treasuries, where, the biggest difference is the 5Y treasury at a staggering 13.022% difference in its average yield. Now, we will transition more to data visualization as the analysis of magnitude continues.

TABLE IV: Scheduled vs Unscheduled Hypothesis Table

Asset	$\mu_{scheduled}$	$SE_{scheduled}$	$\mu_{unscheduled}$	$SE_{unscheduled}$	Difference	T stat	P-value
S&P 500 (%)	-0.008	0.198	-5.32	5.319	5.312	0.998	0.392
Russell 2000 (%)	-0.003	0.257	-7.813	6.448	7.81	1.21	0.313
Financials (%)	-0.047	0.266	-7.172	7.754	7.125	0.918	0.426
Gold (%)	0.028	0.208	-0.903	3.227	0.931	0.288	0.792
Silver (%)	-0.09	0.328	-4.476	7.626	4.385	0.575	0.606
Dollar (%)	-0.102	0.092	0.061	0.554	-0.163	-0.289	0.79
2Y yield (bp)	-2.185	0.827	-9.5	10.836	7.315	0.673	0.549
5Y yield (bp)	-2.842	0.977	-4.75	13.022	1.908	0.146	0.893
10Y yield (bp)	-2.671	1.007	0.25	11.982	-2.921	-0.243	0.824
30Y yield (bp)	-2.062	0.9	4	4.143	-6.062	-1.43	0.24
VIX (pts)	-0.113	0.292	7.637	8.197	-7.75	-0.945	0.414

Lining up all the securities (omitting treasuries) with respect to the event window [-1,1] to observe changes around the average asset return or price change for unscheduled/emergency and scheduled FOMC meetings demonstrates how each asset class reacts. Immediately, in the scheduled sample (n=146), we can observe tiny margins of error for every single security. There is minimal variance explained by the incredibly small standard error. On the other hand, the seldom unscheduled or emergency subsample (n=4) displays the exact opposite. Due to the inherent rarity of the emergency meetings, emergency cuts seem to completely displace market averages and cause large variances. Quantifying the bar chart, further shows that emergency FOMC decisions cause a form of market risk that shakes almost every market, the US dollar evades this volatility for both samples. In the scheduled subsample, the S&P 500, Russell 2000, Financials, precious metals, and the VIX index all reaction efficiently to planned FOMC decisions, shown in the small standard errors and even smaller means. The emergency FOMC meetings have incredibly large and sometimes disproportionate differences with their means, for example, all the 5Y treasury yield has a 1.908 average change in its yield yet a standard error of 13.022. These differences trend for the other treasuries except for the 30Y, being the least sensitivity fixed income security in the study, followed by the US dollar being least affected.

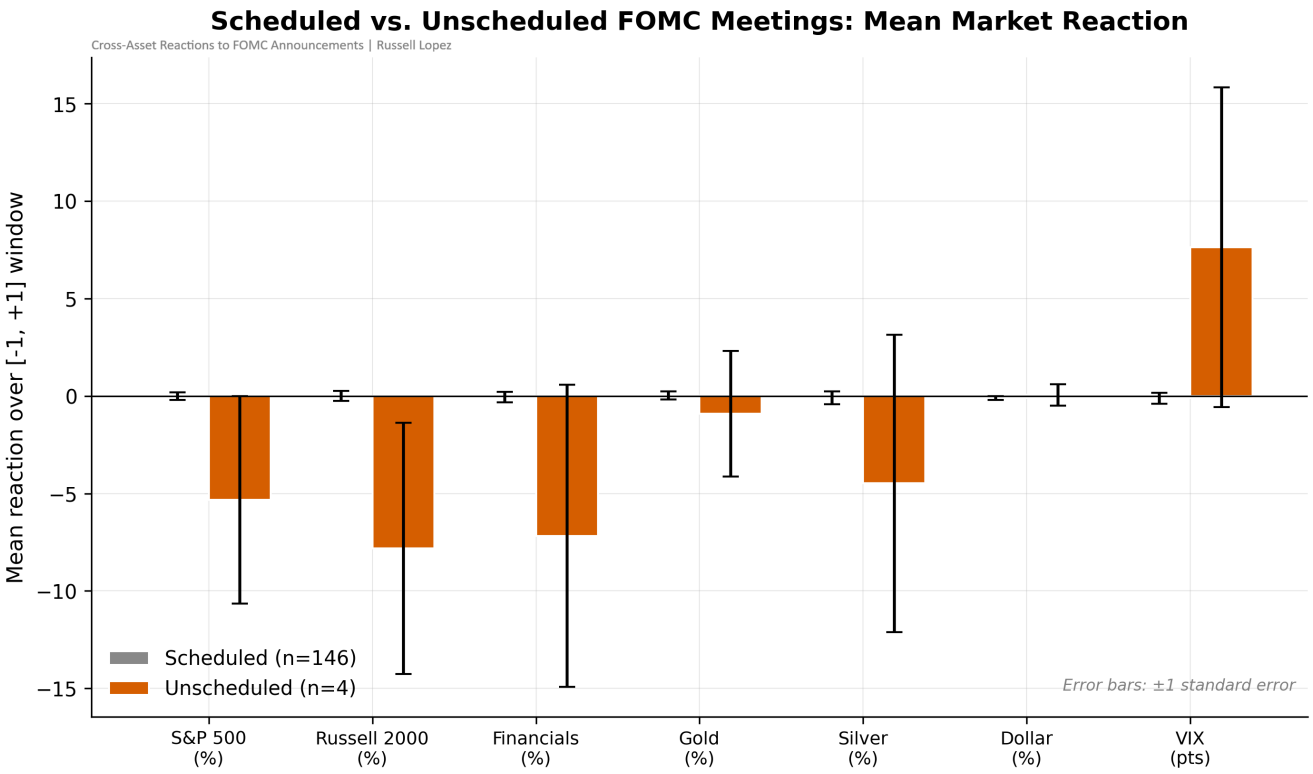


FIG. 3. Scheduled vs Emergency Comparison

The four outlier subsamples demonstrate extreme volatility within the respective event window. Clearly, the Fed unexpected enacts policy rate change, it induces systematic risk to all markets, causing large swings in the S&P 500 and volatility index. Tersely, signaling market panic, fear, and uncertainty (it makes sense as both time periods were unprecedented, after the burst of a global financial bubble and beginning of a worldwide pandemic).

TABLE V. Market Environment Preceding Emergency FOMC Meetings

	Preceding 30 days			**Event window** $[-1, +1]$	
Meeting date	**S&P (%)**	**VIX avg**	**VIX prior day**	**S&P CAR (%)**	**ΔVIX (pts)**
Jan 22, 2008	-9.98	23.2	27.2	+2.13	-1.99
Oct 08, 2008	-22.07	35.5	53.7	-14.64	+11.87
Mar 03, 2020	-3.99	20.3	33.4	+5.50	-8.12
Mar 15, 2020	-21.58	35.4	57.8	-14.27	+28.79

Notes: The four unscheduled FOMC meetings in the 2008–2026 sample. 'S&P (%)' reports the cumulative log return of the S&P 500 over the 30 calendar days prior to the meeting. 'VIX avg' is the mean daily closing VIX over that same window; 'VIX prior day' is the closing VIX on the business day immediately preceding the announcement. Event window statistics ($[-1, +1]$ business days) are reported for direct comparison with the pre-meeting environment.

Diving deeper in the analysis of fixed income sensitivity, let us observe US treasuries in the event window, [-1,1]. Evidently, in the model, US treasuries are by far the most sensitive security to change in the federal fund rate. Figure 4 contrasts the yield differences from short-term treasury yields to long-term, before and after FOMC announcements. In hike meetings (when the FOMC votes in favor to raise policy rates), there is steady rise in yields as the maturity date increases, a natural reaction for treasuries. The variance is minimal and reaction is expected. As for rate cut meetings (when the FOMC votes in favor to cut policy rates), the change is more turbulent with a precipitous rise in yields from 2-year treasuries to 10 years; signaling a low-interest rate environment where debt becomes extremely cheap for all borrowers. For both announcement types, the treasury yield (%) from the pre-announcement (days before) and post-announcement (day after) is minimal, signaling convexity and fixed income efficiency.

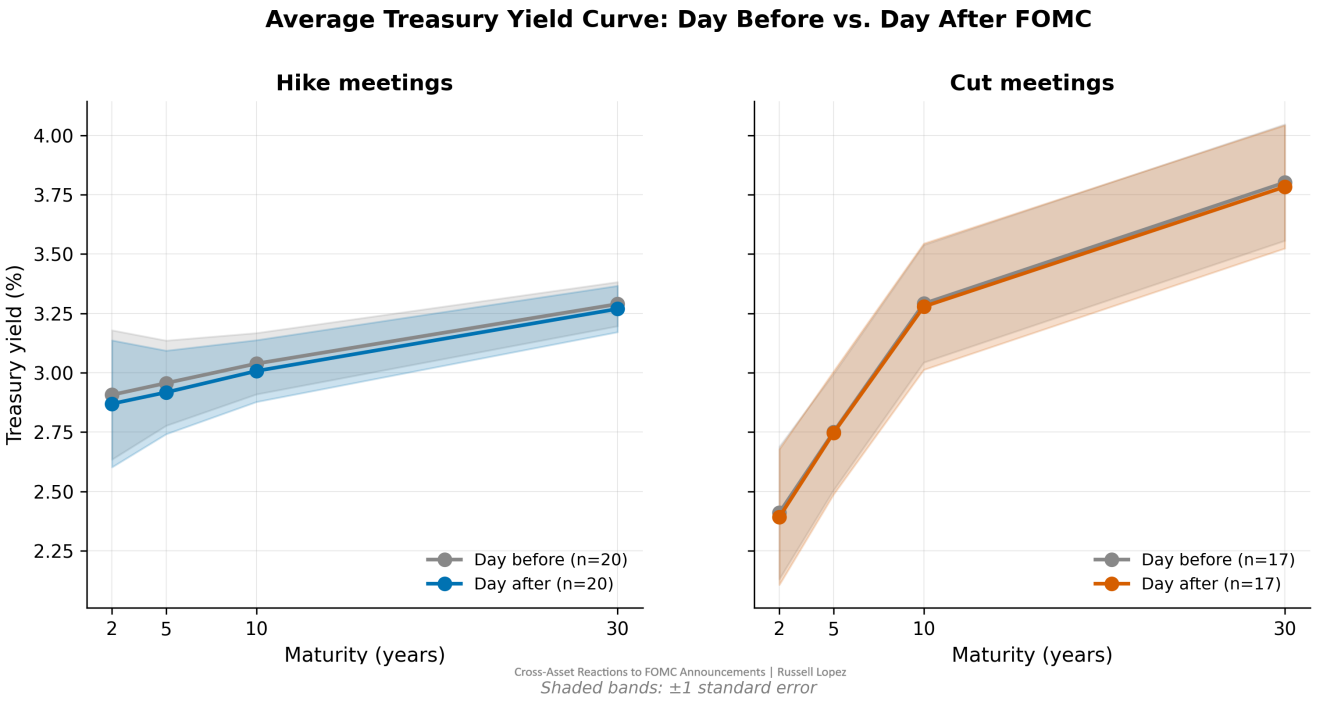


FIG. 4. Yield Curve Shift on Hike vs Cut Days

When quantifying interest rate hike meetings, there is a clear demonstration of marginal growth for yield rates as maturities rise, with the 2Y and 30Y spread being 0.383%, less than half. Such a fractional difference in yields reflects expectations of long-term economic growth to slow, pushing long-term yields. Policy rate hikes tend to have a more positive effect on the 2Y and 5Y yields, higher than the respective yield rates for policy rate hikes. Interestingly, longer term yield rates for the 10Y and 30Y are less, as longer term treasuries are naturally less sensitive to concurrent policy rates. The standard error post-announcement falls as the length of the treasuries rise, signaling efficient mean aversion (the market correcting).

TABLE VI. Figure 4 Table Note for Hike Meetings (Post SE means standard error on the post-meeting value)

Hike meetings (n=20)				
Maturity	Pre (%)	Post (%)	Shift (bp)	Post SE
2Y	2.907	2.870	-3.75	0.267
5Y	2.956	2.917	-3.90	0.176
10Y	3.038	3.007	-3.10	0.130
30Y	3.290	3.269	-2.05	0.097

On the contrary, rate cut meetings show a rather staggering result. The disproportional rate in yields presents not only greater long-term yields overall but a greater spread difference. We can clearly see large jumps in the yield spreads for hike meetings; 0.34bp from 2Y to 5Y, 0.541bp from 5Y to 10Y, and 1.511bp from 10Y to 30Y. Clearly, shorter term treasuries are extremely susceptible to federal fund rate increases, dropping off as the treasuries increase in maturity and this effect accentuates the more aggressive the policy rate hikes. Likewise, the standard error for all treasuries exceeds 0.25, signaling large differences in yields overall.

TABLE VII. Figure 4 Table Note for Cut Meetings (Post SE means standard error on the post-meeting value)

Cut meetings (n=17)				
Maturity	Pre (%)	Post (%)	Shift (bp)	Post SE
2Y	2.410	2.392	-1.76	0.286
5Y	2.750	2.746	-0.35	0.259
10Y	3.291	3.279	-1.24	0.266
30Y	3.802	3.784	-1.82	0.259

For major equities, cumulative abnormal returns will capture market expectations, where the baseline ($CAR(\%) = 0$) in figure 5, signifies an exact and expected performance by the market without any net change in prices or abnormal returns. The closer that CAR consolidates to the baseline, the more efficient market prices tend to be. This is demonstrated extremely well in the distribution of S&P 500 reactions to FOMC decisions, when policy rates are held (gray shaded region), there are minimal outliers and most

of the data is concentrated around the CAR baseline which translates to an efficient market. Profound observations can be seen in unexpected rate cuts in the orange region followed by rate hikes in the blue region. First, beginning with policy rate hikes, most notably on March 16th, 2022, the federal fund rate was raised from 0.25% to 0.50%, yet the CAR climbs to a staggering 5.384%. Several months later, on November 2nd, 2022, the federal fund rate was 3.75% to 4.00%, where, the CAR was down to -3.707%. When rates are hiked, most of the data clumps just below this outlier. Next, observing the most volatile region of the graph is in the center, containing the rate cuts, where we see two incredible outliers on the positive and negative side. First, the former took place on October 29th, 2008, where the federal fund rate was lowered to 1.00%, the latter was slightly behind on October 8th, 2008, where the federal fund rate was marginally higher at 1.50%, yet the CAR cratered to -14.642%. The variance for all the existing CAR reactions is quite spread out from the baseline; illustrating that the FOMC decision to cut policy rates cause the most unpredictable reactions by far. Third and lastly, the least volatile and most concentrated data around the baseline are in the region where rates are held steadily. Once again, there are two notable outliers in the data; on March 18th, 2008, where the federal fund rate was 2.25% and the CAR is 5.032%, whereas on September 16th, 2008, the policy rate was 2.00% with a CAR of -7.791%. All three samples show staggering results with anomalies and do a suffice job in displaying CAR for the S&P 500.

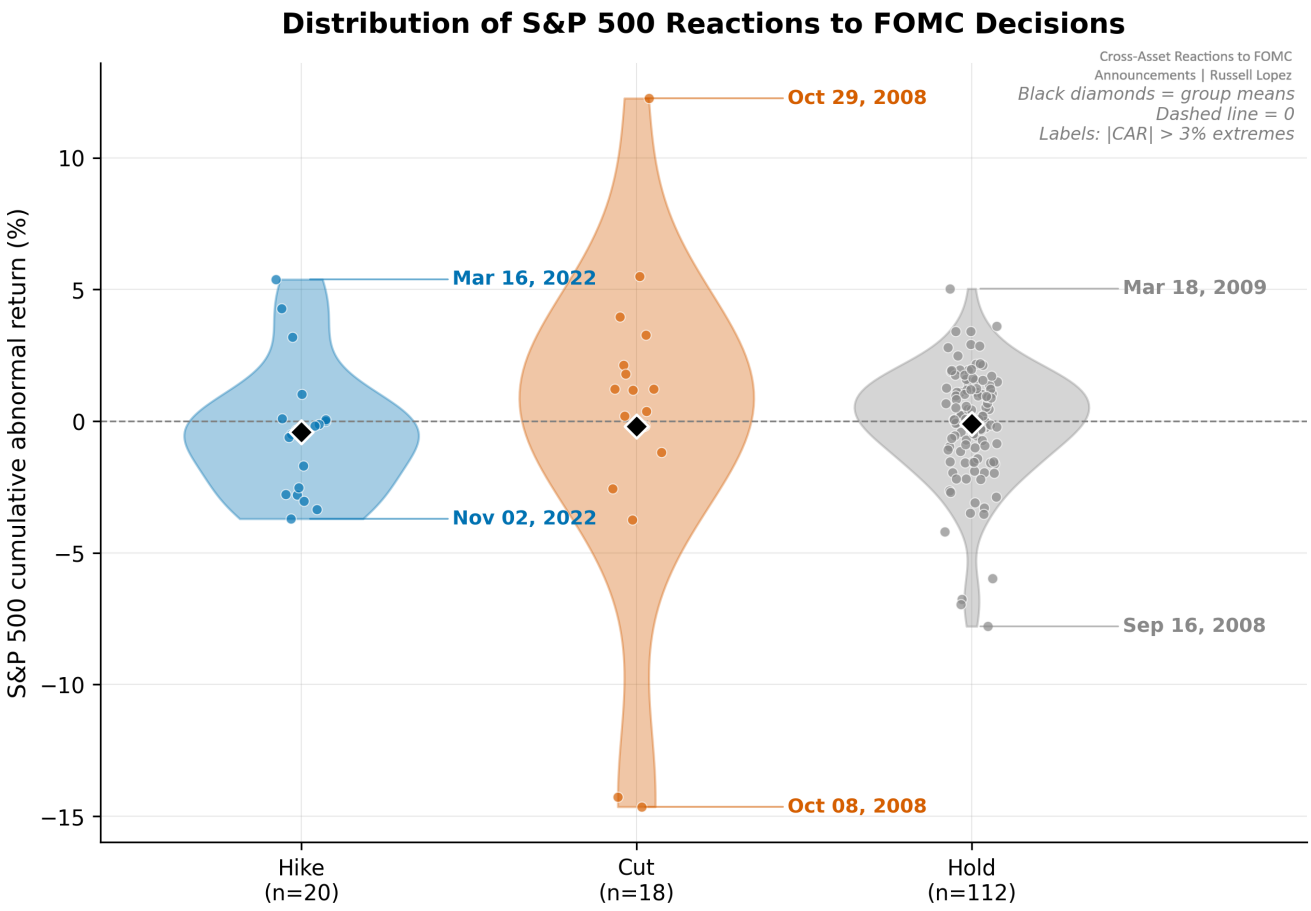


FIG. 5. Distribution of S&P CAR by Decision

Outliers quantified for the distribution figure of S&P 500 by decision. Complementary to figure 5 but it must be restated that these not casual claims but statistical observations, the former would lead to endogeneity bias.

TABLE VIII. Post SE means standard error on the post-meeting value

Decision	Date	Extremum	S&P 500 CAR (%)
Hike	2022-11-02	Most negative	-3.707
Hike	2022-03-16	Most positive	5.384
Cut	2008-10-08	Most negative	-14.642
Cut	2008-10-29	Most positive	12.259
Hold	2008-09-16	Most negative	-7.791
Hold	2009-03-18	Most positive	5.032

A central finding for the event study was the S&P 500 reacts to the three major FOMC decisions (hike, cut, and hold) across different Fed regimes. A harsher and darker color for each cell entails a higher (red) or lower (blue) cumulative average return over the event window, [-1,1]. Unfortunately, due to the limited scope of subsamples, not every cell in the matrix could be filled. In the heatmap, there is general propensity for the S&P 500 to react negatively across each FOMC decision in each subsample. The most drastic negative CAR can be observed in the COVID-19 sample when policy rates are cut, resulting in a -4.38%, not a casual finding, rather a coincidental finding. The highest CAR is in that same FOMC decision to cut rates, but, instead in the GFC subsample, at 1.13%. The most complete data comes in the form of holding rates (due to the most abundant amount of data), were, we can see minor changes in the CARs, with the irregularity spotted in GFC subsample (CAR of 2.65%). Overall, limited but insightful visualization for the S&P 500.

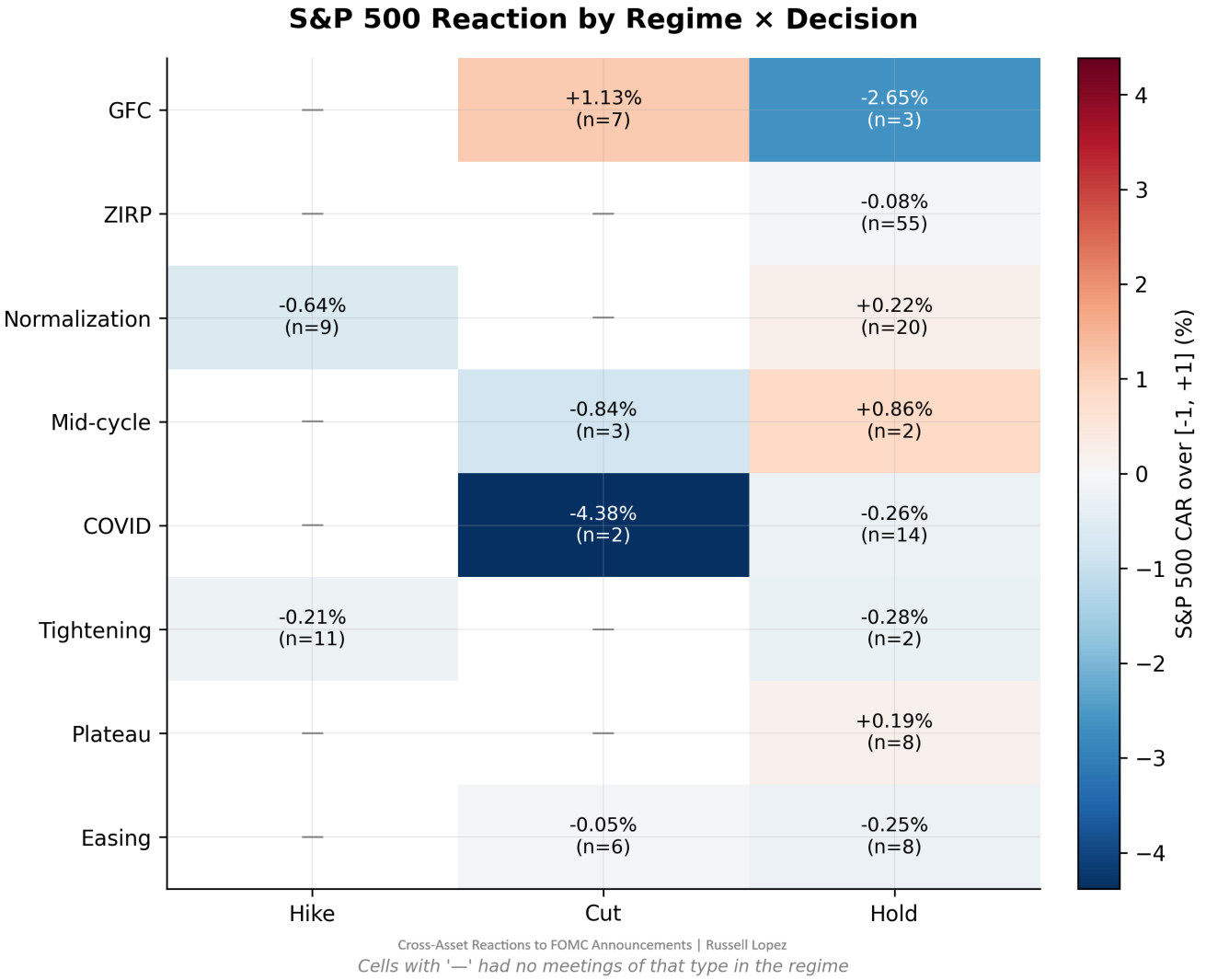


FIG. 6. Heatmap: S&P 500 Reaction by Regime × Decision

The CBOE Volatility Index is an valuable benchmark to gauge market fear and uncertain anticipation, with the higher the index, the greater fear and panic in the market. In addition, this index acts as a proxy for observatory behavior in the derivatives market, more specifically, the mid-quote prices in the options market. The VIX behavior graph in figure 7 encapsulates the cumulative change in the beginning five days before the official FOMC announcement (at x=0), followed by five days after. The greater shaded region signals greater standard error and more volatility. The most conspicuous finding is by far when the federal fund rate is cut, the shaded bands are overwhelmingly large. Now, it is very important to note

that the primary factor for this variance is due to low statistical power in the relatively small samples for hikes (n=18) and cuts (n=14), therefore, we can not make any statistical inferences. Nonetheless, the exploratory findings show that for rate cuts; there is big jumps in the cumulative change in the VIX index, signaling high stress in the market regardless of what the Fed enacts. This trend of market uncertainty continues until announcement day, and grows larger, post-announcement day. On the contrary, policy rate hikes seem to trend the other way, on the negative change, signaling volatility in the other direction but once again, without referring any casual inferences. In closing, the policy rate holds provide the largest sample (n=105) and thus provide the most reliable result with the volatility index consolidating around the baseline with minor deviations. It seems Fed executes policy in an existing stressed market.

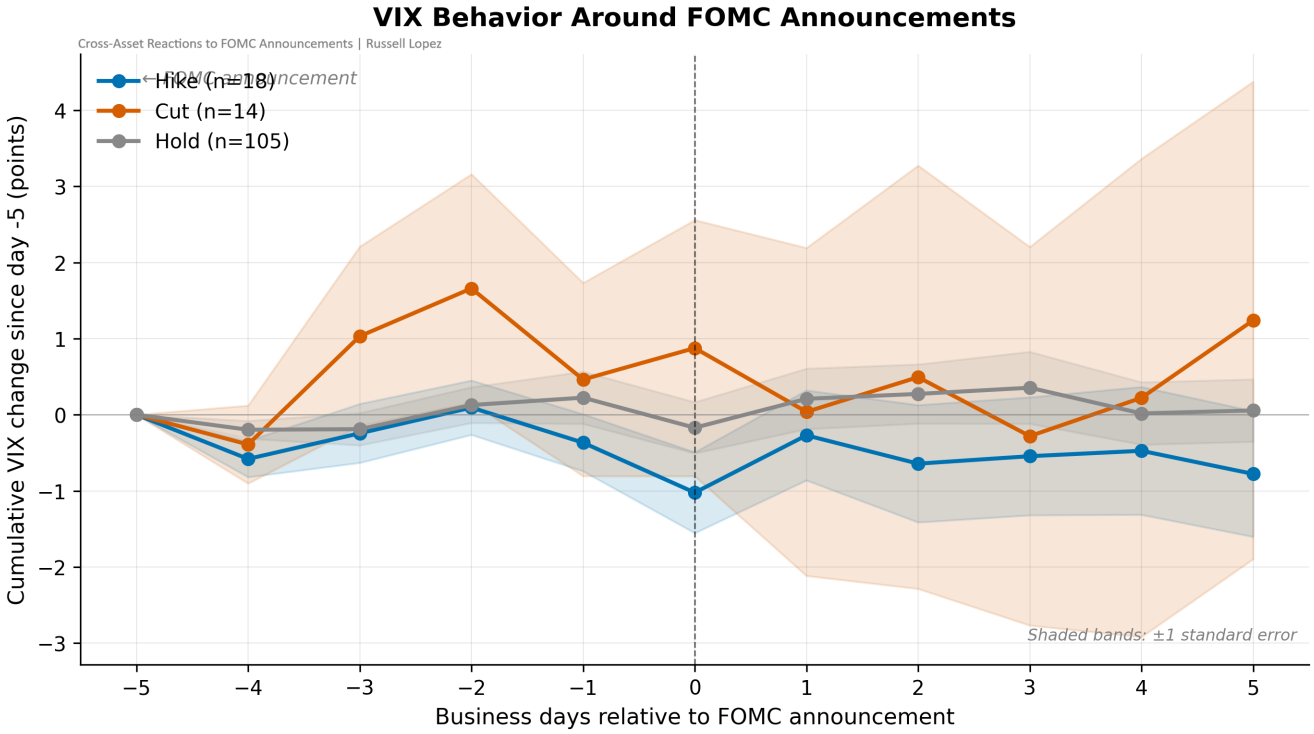


FIG. 7. VIX Behavior Around FOMC Days

Identical to the figure 7's setup, in figure 8, the analysis is on the focus of the S&P 500 for the security of interest. Labeled with the same 11-day event window, [-5.5], for the x-axis and a cumulative change in return rather than CAR as a measure in the y-axis. From the one sample t-test, it is clear that the S&P 500 95% confidence intervals stretch out greatly and include zero, rendering statistical insignificance

meaning that FOMC announcements do make a significant difference. This means no solid casual claim can be made. With that said, the rate cuts produce an extremely wide standard error band, followed by rate hikes, and the tightest spread is when rates are held steady. Evidently, the S&P 500 has been already reacting to systematic risk and official FOMC decision to alter the federal fund rate seems to have no effect or casual link to volatility. The volatility has already been priced in, giving more credence to the strong form efficiency.

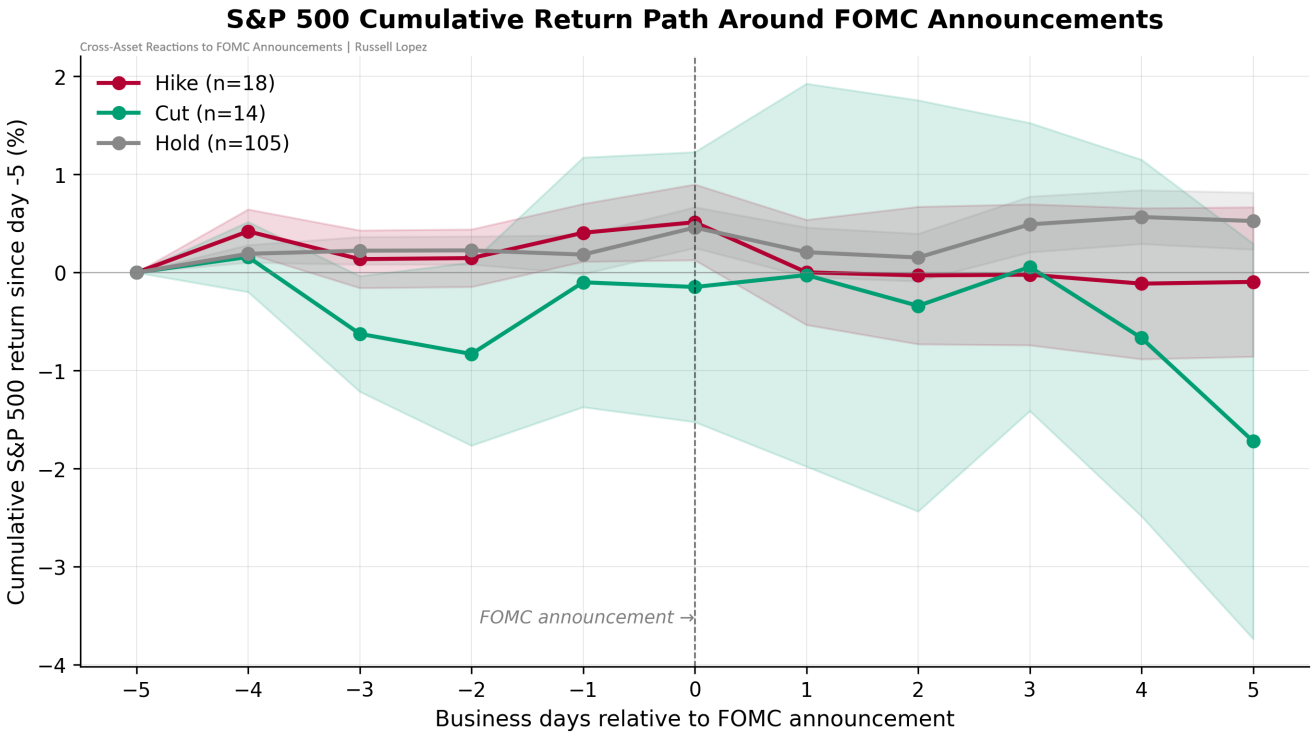


FIG. 8. Cumulative S&P Returns Event-Time

Analysis of cross-asset reactions to FOMC meetings is the main focuses of the study and can be visualized in figure 9 with the conventional 3-day event window, [-1,1], and with a baselines of zero. Recall from the hypothesis t-testing, a 95% confidence intervals that include zero are not statistically significant and show no substantial change in their respective averages, so any asset with a confidence interval that excludes zero are statistically significant. The first graph displays information for policy rate hikes with a modest sample size (n=20); it is evident that no asset's confidence interval excludes zero and all share high within-group variance, this is true for the rate cut sample (n=18), where, all confidence

intervals cross the zero baseline. The last graph displays the strongest sample (n=112), when rates are held and it displays the significant findings for treasuries except for one. There is clear negative change in yields for all four of the fixed income securities (2Y, 5Y, 10Y, 30Y) but only the 2Y, 5Y, and 10Y treasuries have inferential findings, where the treasury yields receive downwards pressure when the Fed holds rates steady. The 30Y 95% confidence interval does cross zero which renders it insignificant to our findings. Concluding the results, it is time to discuss the theory and begin explaining the mechanisms of the market movements, the influence of the Federal Reserve, and asset reactions.

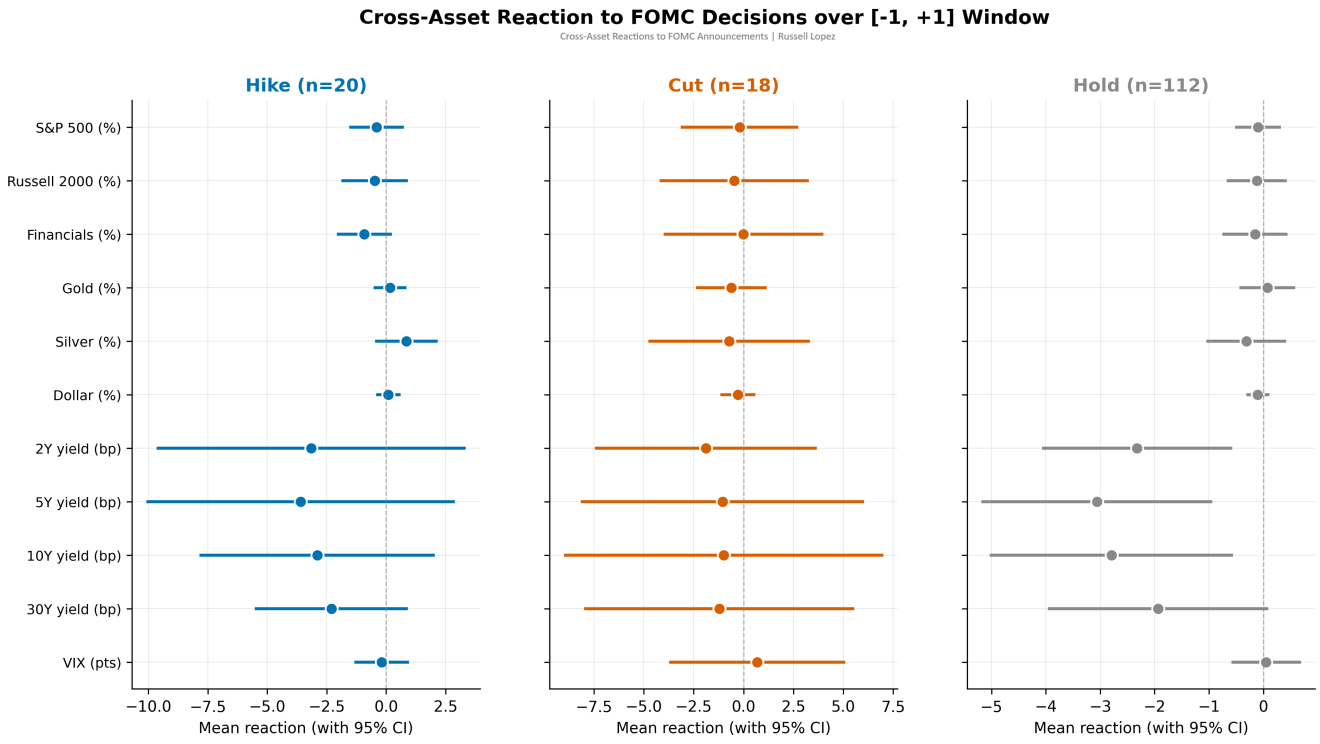


FIG. 9. Cross-Asset Reaction Summary (view Appendix to view complementary descriptive statistics)

TABLE IX. Summary Table of Descriptive Statistics

Variable	***N***	**Mean**	**SD**	**Min**	**Max**	**Skewness**
S&P 500 CAR (%)	150	-0.150	2.931	-14.642	+12.259	-1.223
Russell 2000 CAR (%)	150	-0.211	3.786	-20.137	+13.986	-1.454
Financials CAR (%)	150	-0.237	4.026	-25.685	+12.288	-2.008
Gold return (%)	150	+0.003	2.644	-9.953	+10.752	-0.208
Silver return (%)	150	-0.207	4.528	-26.989	+11.036	-1.472
Dollar return (%)	150	-0.098	1.103	-5.821	+2.388	-1.219
2Y yield change (bp)	150	-2.380	10.388	-59.000	+22.000	-1.647
5Y yield change (bp)	150	-2.893	12.220	-45.000	+34.000	-0.218
10Y yield change (bp)	150	-2.593	12.486	-41.000	+36.000	-0.235
30Y yield change (bp)	150	-1.900	10.839	-44.000	+25.000	-0.620
VIX change (pts)	150	+0.094	4.366	-17.160	+28.790	+2.077

Notes: Descriptive statistics for the 11 event-window reaction variables across all 150 FOMC announcements in the sample (January 2008 – April 2026), measured over the $[-1, +1]$ business-day window. Equity reactions reported as cumulative abnormal returns (CAR) using the constant-mean model; other reactions reported as raw cumulative changes. Skewness computed using the Fisher-Pearson coefficient.

IV. DISCUSSION

In the rigorous statistical testing in differing Fed regimes from 2008 to 2026, the most critical finding was the evidential discovery that the fixed-income market react in the most measurable way to FOMC announcements, specifically when policy rates are held steady, leading treasury yields (2Y, 5Y, 10Y) to see a significant decline in their cumulative yields. The mechanism of fixed-income reaction with relation to monetary policy is defined by convexity, the relationship between short-term interest rates and shorter term treasuries. The secondary findings clearly shows that the equity indices demonstrate no statistically significant results in the pooled sample, or any of the subsamples for that matter. This reinforces the well-known efficient market hypothesis in finance theory, where, market prices already reflect all current public information and are actively forward looking (anticipating future economic conditions), this is why equities are a leading indicator. The last major finding was the confirmation that unscheduled FOMC announcements lead to incredible volatile movements in the markets: -5.32% S&P 500, -7.813% Russell 2000, -7.172% Financials, -4.476% Silver, -9.5bp 2Y, -4.75bp 5Y, +4bp 30Y, and +7.613pts VIX.

A. Treasury Yields and Policy Rate Holds

It is profoundly interesting to observe that the bond market was the only market that had a statistically significant change to any decision type. Recall, in the full sample (n=112) when policy rates are held steady, the 2Y, 5Y, and 10Y treasuries have a average change of -2.321bp, -3.062bp, and -2.795bp respectively. The 30Y just barely escapes the significance threshold as its p-value just barely exceed 0.05 (at p=0.06). Excluding the 30Y, the treasuries fall significantly during hold meetings even when no rate change occurs, this finding aligns with the conventional wisdom; that short-term interest rates are directly influenced by the federal fund rate, leading to fluctuations in fixed-income prices and yields. Treasury yields are computed with three primary variables; the expected real short-term rate (change in the current federal fund rate), expected inflation rate (drives up inflation premium), and the term premium (longer maturities require higher yields). Treasury yields are inversely correlated with their respective prices, acting as opposing forces. Furthermore, the fixed-income market always prices-in future expectations about the economy, it is a forward-looking market. Therefore, any decision the Fed decides to announce will already be priced in. This is why there is no significant changes in our hypothesis testing when policy rates are cut or hiked. But, when rates are held steady, this does invoke a volatile reaction, and this is because of two possible scenarios. First, if the fixed-income market is taken by surprise with expectation of an imminent policy rate hike (or rate cut) but instead are met with a policy rate hold, this will drive investors to purchase treasuries as the threat of a uncertainty grows, such rapid demand for treasuries puts upwards pressure on their prices, decrease yields. The 30Y treasury is typically immune to this response as the maturity date is so far into the future that investor action now is negligible. The second scenario is the forward guidance effect, where, even if the fixed-income market fully expects the rate holds and guesses correctly, there will still be an influx of volatility because the Fed's decision to hold rates only confirms greater uncertainty about the future. Holding the policy rate steady typically indicates that interest rates will continue to be held at their current rate or even increased further, a scenario that happens as measure to combat high inflation.

B. Equity Reaction Findings

Cumulative average returns were an exclusive measure for equities in the study, as such, it embedded not only the net change in cumulative return but also less the expected return. As a result, if market

expectations of returns were equal to actual returns, it would equate to a CAR of zero, signifying expectations being totally aligned with the outcome. Any deviations from the baseline of zero in the positive or negative direction signaled the market was caught by surprise, resulting in abnormal gains or losses. With that said, recall to the table 11 (summary table of descriptive statistics), where in the full sample (n=150) for the 'all' decision (average of hike, rate, and cut), the CAR results as listed: S&P 500 CAR of -0.150%, Russell 2000 CAR of -0.211%, and Financials CAR of -0.237%. This indicates that despite the extreme outliers and variance in figure 5 (distribution of S&P 500 reactions to FOMC meetings), the average CARs for all three major equity indices across our entire sample experience minimal market shock. This finding coincides with the efficient market theorem explained by the pricing-in mechanism, where the equity market evidently already foresees what the FOMC committee will announce before the official announcement is made at 2PM. This is consistent with the small averages for each of the respective equity indices, if it was the case that the opposite was true, the averages would be substantially deviate from zero. Further, the findings suggest that in the long-term despite emergency policy announcements, abnormal returns are rare and the forward looking stock market's future projections of economic outlook aligns with actual contingent economic outcomes.

C. The Endogeneity of Fed Action to Market Conditions

In reference to table 6 (scheduled vs unscheduled meetings table), the most profound finding is differences in the averages for every asset class for scheduled versus emergency FOMC meetings, with exception of the US dollar. Illustrating a classical concept in econometrics known as endogeneity or reserve causality. The initial question was to analyze how the markets react to Fed policy rate changes, when in fact, the findings show that the Fed decides the policy rate based on how the markets already reacted. This is not to say for instance, that the stock market causes the Fed to enact certain policy decisions but in fact both entities are reacting to an existing economic crisis. Emergency meetings are exceptionally rare, in the entire sample there is only four, two in 2008; a 75bp rate cut on January 22nd and a rate cut in October 8th, the other two were in 2020; a rate cut of 50bp in March 3rd followed by a 100bp cut on March 15th. In these subsamples, the four outliers were in experiencing pre-existing economic pressures and market shocks that forced the Fed to act, highlighting endogeneity in its purest form. In the first outlier, on Jan.22nd, 2008, the subprime mortgage crisis continued to escalate after the burst of the housing bubble,

and the S&P 500 was down -9.98% in the last 30 days, ushering an emergency decision to slash rates by 75 basis points. Global equity markets were cratering, the housing market was contracting, and credit markets were being squeezed, as a result, the Fed panicked and was forced to cut policy rates quickly and in spite of already hemorrhaging asset markets. On Oct. 8th, 2008, numerous months after the first initial emergency meeting, there was another effort to slash policy rates but this time in a coordinated global effort to cut rates by 50 basis points. At this point, the global financial crisis had cascaded worldwide, and therefore central banks (ECB, BoE, etc.) from other major economies collaborated to simultaneously to also cut their respective policy rates by 50bp. Illustrating a prime case of reverse causality, where the stock market became a predictor variable and change in policy rate became the response variable. This concept is reinforced in the other half of the emergency meeting subsample. Where in March 3rd of 2020, emergency rates were cut by 50 basis points, the first ever COVID-19 cut. The S&P 500 was down -3.99% from the prior thirty days. 12 days later, the final emergency meeting was held on a Sunday on March 15th, where there was a rate cut of 100 basis points, resulting in a zero-interest rate environment. The S&P 500 plummeted to -21.58% as sentiment fears of global lockdowns grew as the pandemic spread. Lastly, it is worth mentioning that the VIX index shot up to +28.79 the following day (and an all-time closing high of 82.69 on March 16), confirming heavy anticipations of price swings and immense market unpredictability for the future.

D. Limitations of The Study

The scope of this event study has its limitations which ultimately impacts statistical testing and the inferences that can be made, there are three main concerns. First, due to the nature of seldom emergency meetings, any formal inferences can not be made between the mean differences among the scheduled and unscheduled data. Therefore, explanations for extreme events can only be made from qualitative observation. Future research should include a greater subsample of outliers. Second, the emphasis on the policy rate change omitted other monetary policy tools that the Fed carries in its arsenal such as expanding or contrasting its balance sheet and forward guidance communication. Both are effective tools used to manipulate the money supply and spread market sentiment which ultimately acts as new information introduced into the markets. For instance, if the Fed decided to expand their balance sheet, this would come in the form of purchasing debt securities from the open markets, subsequently, growing the money

supply. An increase in the money supply would have second order effects on the greater markets. For forward guidance, this can be communicated in press conferences where the corresponding Fed chairman states explicit keywords and statements regarding projected outcomes for the economy. Adding these variables to our study would require much more extraneous testing and would be hard to quantify but would reduce excess noise (non-significant change). Third, the exclusion of a surprise decomposition such as the federal fund futures slightly reduced the quality of the findings, as the futures market are key in predicting future metrics such as the average federal fund rate. Taking into account a surprise decomposition like fed funds futures would provide a clear observation of what the market expected in form of derivative contracts. In this paper, the focus was measuring market sensitivity following realized decisions, this makes it truly difficult to differentiate what was real reactions were and what was already priced in.

E. Market Implications and Notes For Future Research

The strongest actionable finding in this study for finance practitioners and investors alike is to pay close attention to fixed-income markets when no-change FOMC meetings are anticipated. Because of the systematic shock to treasuries when policy rates are held steady, the yield curve will flatten (until the 10Y maturity treasuries), signaling forward guidance. Also, in times of market uncertainty, the equity markets see huge downswings, discounting stocks that are intrinsically and fundamentally valuable. Capitalizing on the market's overreaction when systematic shocks bleed the markets may prove to work in the investor's favor, if the correct companies are purchased. The same principle can apply inversely, when markets rally to the extreme, causing cumulative average returns to abnormally soar beyond their fair share prices, shorting can be a viable yet risky trade to make as market prices return to their normal levels. On the literature frontier, the work of this paper proves the limitations of a realized-decision based framework, thus, successor researchers should attempt to implement predictor indicators like the CME FedWatch tool which predicts the change in the federal fund rates based on a complex probability model. Including this type of scaffolding in future research would provide higher quality insight into market expectations and price discrepancies.

V. CONCLUSION

It is clear that the Federal Reserve's power over the capital markets is influential in changing market direction completely. Evidently, the broader capital markets operate as forward looking vehicles that continuously project economic outlooks reflective in their spot prices. When the Fed acts on monetary policy, this injects new information into the system and either concurs or alters market sentiment, the reverse is found to true, if markets are reflecting underlying, unfavorable economic conditions, the Fed will act on it. In times of such economic turbulence, volatility is inevitable in the buildup, press conference announcement, and the days after. The lingering effects of volatility are shaped with all publicly available information which the markets use as guidance. Sometimes, the market gets it wrong, producing mispricing phenomenons, opening the window of opportunity for prudent yet risk-seeking investors. For the risk-averse, in times uncertainty, hedging in efficient asset classes can protect the value of wealth. Understanding market reactions can be helpful in forming exceptions on what is to come and studying shocks is vital to comprehend upcoming economic cycles, deviations in baseline prices and greater investor confidence.

VI. APPENDIX

TABLE X: Welch's Two Sided T-Test For Hypothesis Testing of Contrasting Policy Decisions For Each Asset

Sample	Contrast (a vs b)	Metric	μ_a	μ_b	Diff.	T Stat.	P-value	Sig.	N_a	N_b
Full sample	hike vs cut	S&P 500 CAR (%)	-0.404	-0.201	-0.202	-0.129	0.898	–	20	18
Full sample	hike vs cut	Russell 2000 CAR (%)	-0.486	-0.474	-0.013	-0.006	0.995	–	20	18
Full sample	hike vs cut	Financials CAR (%)	-0.917	-0.010	-0.907	-0.436	0.667	–	20	18
Full sample	hike vs cut	Gold (%)	0.159	-0.608	0.767	0.832	0.415	–	20	18
Full sample	hike vs cut	Silver (%)	0.848	-0.726	1.573	0.740	0.468	–	20	18
Full sample	hike vs cut	Dollar (%)	0.089	-0.284	0.374	0.798	0.432	–	20	18
Full sample	hike vs cut	2Y yield (bp)	-3.150	-1.889	-1.261	-0.293	0.772	–	20	18
Full sample	hike vs cut	5Y yield (bp)	-3.600	-1.056	-2.544	-0.524	0.603	–	20	18
Full sample	hike vs cut	10Y yield (bp)	-2.900	-1.000	-1.900	-0.400	0.692	–	20	18
Full sample	hike vs cut	30Y yield (bp)	-2.300	-1.222	-1.078	-0.286	0.778	–	20	18
Full sample	hike vs cut	VIX (pts)	-0.189	0.677	-0.865	-0.379	0.709	–	20	18
Full sample	hike vs hold	S&P 500 CAR (%)	-0.404	-0.096	-0.308	-0.525	0.605	–	20	112
Full sample	hike vs hold	Russell 2000 CAR (%)	-0.486	-0.120	-0.366	-0.502	0.620	–	20	112
Full sample	hike vs hold	Financials CAR (%)	-0.917	-0.152	-0.765	-1.212	0.235	–	20	112
Full sample	hike vs hold	Gold (%)	0.159	0.074	0.085	0.211	0.834	–	20	112
Full sample	hike vs hold	Silver (%)	0.848	-0.313	1.160	1.577	0.125	–	20	112
Full sample	hike vs hold	Dollar (%)	0.089	-0.101	0.191	0.766	0.451	–	20	112
Full sample	hike vs hold	2Y yield (bp)	-3.150	-2.321	-0.829	-0.244	0.810	–	20	112

Continued on next page

TABLE X: Welch's Two Sided T-Test For Hypothesis Testing of Contrasting Policy Decisions For Each Asset

Sample	Contrast (a vs b)	Metric	μ_a	μ_b	Diff.	T Stat.	P-value	Sig.	N_a	N_b
Full sample	hike vs hold	5Y yield (bp)	-3.600	-3.062	-0.537	-0.156	0.877	–	20	112
Full sample	hike vs hold	10Y yield (bp)	-2.900	-2.795	-0.105	-0.038	0.970	–	20	112
Full sample	hike vs hold	30Y yield (bp)	-2.300	-1.938	-0.362	-0.190	0.850	–	20	112
Full sample	hike vs hold	VIX (pts)	-0.189	0.051	-0.239	-0.375	0.710	–	20	112
Full sample	cut vs hold	S&P 500 CAR (%)	-0.201	-0.096	-0.105	-0.071	0.944	–	18	112
Full sample	cut vs hold	Russell 2000 CAR (%)	-0.474	-0.120	-0.354	-0.187	0.854	–	18	112
Full sample	cut vs hold	Financials CAR (%)	-0.010	-0.152	0.142	0.070	0.945	–	18	112
Full sample	cut vs hold	Gold (%)	-0.608	0.074	-0.682	-0.758	0.458	–	18	112
Full sample	cut vs hold	Silver (%)	-0.726	-0.313	-0.413	-0.201	0.843	–	18	112
Full sample	cut vs hold	Dollar (%)	-0.284	-0.101	-0.183	-0.438	0.666	–	18	112
Full sample	cut vs hold	2Y yield (bp)	-1.889	-2.321	0.433	0.148	0.884	–	18	112
Full sample	cut vs hold	5Y yield (bp)	-1.056	-3.062	2.007	0.537	0.597	–	18	112
Full sample	cut vs hold	10Y yield (bp)	-1.000	-2.795	1.795	0.427	0.674	–	18	112
Full sample	cut vs hold	30Y yield (bp)	-1.222	-1.938	0.715	0.201	0.843	–	18	112
Full sample	cut vs hold	VIX (pts)	0.677	0.051	0.626	0.280	0.783	–	18	112
GFC	cut vs hold	S&P 500 CAR (%)	1.134	-2.649	3.783	0.894	0.405	–	7	3
GFC	cut vs hold	Russell 2000 CAR (%)	1.136	-2.651	3.786	0.885	0.402	–	7	3
GFC	cut vs hold	Financials CAR (%)	1.565	-3.707	5.271	0.786	0.462	–	7	3
GFC	cut vs hold	Gold (%)	-0.621	3.385	-4.006	-0.902	0.445	–	7	3
GFC	cut vs hold	Silver (%)	0.934	1.329	-0.395	-0.080	0.940	–	7	3
GFC	cut vs hold	Dollar (%)	-1.193	-0.429	-0.764	-0.650	0.537	–	7	3
GFC	cut vs hold	2Y yield (bp)	-0.714	-28.000	27.286	1.413	0.275	–	7	3
GFC	cut vs hold	5Y yield (bp)	0.286	-19.667	19.952	1.186	0.319	–	7	3
GFC	cut vs hold	10Y yield (bp)	-0.857	-12.000	11.143	0.727	0.503	–	7	3
GFC	cut vs hold	30Y yield (bp)	-4.000	-7.333	3.333	0.263	0.804	–	7	3
GFC	cut vs hold	VIX (pts)	-2.197	3.170	-5.367	-1.074	0.333	–	7	3
Normalization	hike vs hold	S&P 500 CAR (%)	-0.637	0.220	-0.857	-1.644	0.124	–	9	20
Normalization	hike vs hold	Russell 2000 CAR (%)	-0.667	0.113	-0.781	-1.291	0.219	–	9	20
Normalization	hike vs hold	Financials CAR (%)	-1.085	-0.132	-0.953	-1.406	0.187	–	9	20
Normalization	hike vs hold	Gold (%)	-0.194	0.647	-0.840	-1.315	0.209	–	9	20
Normalization	hike vs hold	Silver (%)	-0.307	1.004	-1.311	-1.225	0.243	–	9	20
Normalization	hike vs hold	Dollar (%)	0.419	-0.325	0.744	1.820	0.096	*	9	20
Normalization	hike vs hold	2Y yield (bp)	1.556	-2.100	3.656	1.560	0.139	–	9	20
Normalization	hike vs hold	5Y yield (bp)	1.000	-3.150	4.150	1.354	0.201	–	9	20
Normalization	hike vs hold	10Y yield (bp)	-2.111	-2.350	0.239	0.107	0.916	–	9	20
Normalization	hike vs hold	30Y yield (bp)	-4.556	-1.800	-2.756	-1.871	0.074	*	9	20
Normalization	hike vs hold	VIX (pts)	0.552	-0.468	1.020	1.123	0.283	–	9	20
Mid-cycle	cut vs hold	S&P 500 CAR (%)	-0.841	0.859	-1.700	-1.970	0.187	–	3	2
Mid-cycle	cut vs hold	Russell 2000 CAR (%)	-1.068	0.307	-1.375	-2.643	0.135	–	3	2
Mid-cycle	cut vs hold	Financials CAR (%)	-1.285	1.593	-2.878	-2.713	0.107	–	3	2
Mid-cycle	cut vs hold	Gold (%)	0.401	0.465	-0.063	-0.120	0.915	–	3	2
Mid-cycle	cut vs hold	Silver (%)	-0.378	0.804	-1.182	-0.893	0.464	–	3	2
Mid-cycle	cut vs hold	Dollar (%)	-0.150	-0.169	0.019	0.072	0.948	–	3	2
Mid-cycle	cut vs hold	2Y yield (bp)	-8.000	0.000	-8.000	-1.600	0.208	–	3	2
Mid-cycle	cut vs hold	5Y yield (bp)	-11.333	0.500	-11.833	-1.713	0.221	–	3	2
Mid-cycle	cut vs hold	10Y yield (bp)	-12.333	1.500	-13.833	-2.093	0.178	–	3	2
Mid-cycle	**cut vs hold**	**30Y yield (bp)**	**-13.667**	**2.000**	**-15.667**	**-4.075**	**0.046**	******	**3**	**2**
Mid-cycle	cut vs hold	VIX (pts)	1.510	-2.330	3.840	2.105	0.158	–	3	2
COVID	cut vs hold	S&P 500 CAR (%)	-4.384	-0.255	-4.129	-0.417	0.748	–	2	14
COVID	cut vs hold	Russell 2000 CAR (%)	-8.343	-0.734	-7.610	-0.643	0.635	–	2	14
COVID	cut vs hold	Financials CAR (%)	-5.005	-0.702	-4.303	-0.466	0.721	–	2	14
COVID	cut vs hold	Gold (%)	-2.574	0.049	-2.623	-0.355	0.783	–	2	14
COVID	cut vs hold	Silver (%)	-11.111	0.253	-11.364	-0.714	0.604	–	2	14
COVID	cut vs hold	Dollar (%)	0.408	-0.253	0.661	0.536	0.683	–	2	14

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TABLE X: Welch's Two Sided T-Test For Hypothesis Testing of Contrasting Policy Decisions For Each Asset

Sample	Contrast (a vs b)	Metric	μ_a	μ_b	Diff.	T Stat.	P-value	Sig.	N_a	N_b
COVID	cut vs hold	2Y yield (bp)	-16.500	-1.143	-15.357	-5.691	0.066	*	2	14
COVID	**cut vs hold**	**5Y yield (bp)**	**-15.500**	**-1.000**	**-14.500**	**-6.250**	**0.002**	*******	**2**	**14**
COVID	**cut vs hold**	**10Y yield (bp)**	**-10.000**	**-0.643**	**-9.357**	**-3.897**	**0.002**	*******	**2**	**14**
COVID	cut vs hold	30Y yield (bp)	3.000	-0.643	3.643	1.469	0.167	–	2	14
COVID	cut vs hold	VIX (pts)	10.335	0.373	9.962	0.538	0.685	–	2	14
Tightening	hike vs hold	S&P 500 CAR (%)	-0.213	-0.281	0.069	0.032	0.978	–	11	2
Tightening	hike vs hold	Russell 2000 CAR (%)	-0.338	-2.282	1.944	0.602	0.633	–	11	2
Tightening	hike vs hold	Financials CAR (%)	-0.780	0.578	-1.358	-1.003	0.387	–	11	2
Tightening	hike vs hold	Gold (%)	0.447	-1.268	1.715	1.188	0.426	–	11	2
Tightening	hike vs hold	Silver (%)	1.792	-2.611	4.403	1.869	0.270	–	11	2
Tightening	hike vs hold	Dollar (%)	-0.180	-0.040	-0.140	-0.096	0.938	–	11	2
Tightening	hike vs hold	2Y yield (bp)	-7.000	13.000	-20.000	-2.435	0.086	*	11	2
Tightening	hike vs hold	5Y yield (bp)	-7.364	7.500	-14.864	-1.933	0.135	–	11	2
Tightening	hike vs hold	10Y yield (bp)	-3.545	2.500	-6.045	-1.082	0.327	–	11	2
Tightening	hike vs hold	30Y yield (bp)	-0.455	-1.500	1.045	0.375	0.715	–	11	2
Tightening	hike vs hold	VIX (pts)	-0.795	0.040	-0.835	-0.899	0.403	–	11	2
Easing	cut vs hold	S&P 500 CAR (%)	-0.045	-0.247	0.202	0.201	0.847	–	6	8
Easing	cut vs hold	Russell 2000 CAR (%)	0.569	-0.072	0.641	0.346	0.742	–	6	8
Easing	cut vs hold	Financials CAR (%)	0.456	-0.006	0.462	0.348	0.740	–	6	8
Easing	cut vs hold	Gold (%)	-0.443	-0.205	-0.238	-0.157	0.878	–	6	8
Easing	cut vs hold	Silver (%)	0.625	-1.436	2.061	0.680	0.511	–	6	8
Easing	cut vs hold	Dollar (%)	0.478	0.291	0.187	0.405	0.694	–	6	8
Easing	cut vs hold	2Y yield (bp)	4.667	1.875	2.792	0.776	0.453	–	6	8
Easing	cut vs hold	5Y yield (bp)	7.333	0.250	7.083	1.929	0.082	*	6	8
Easing	**cut vs hold**	**10Y yield (bp)**	**7.500**	**-1.375**	**8.875**	**2.668**	**0.028**	******	**6**	**8**
Easing	**cut vs hold**	**30Y yield (bp)**	**6.833**	**-1.750**	**8.583**	**2.913**	**0.016**	******	**6**	**8**
Easing	cut vs hold	VIX (pts)	0.393	-0.133	0.526	0.253	0.810	–	6	8

[Note1] [Note2] [Note3]

TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
Full sample	hike	S&P 500 CAR (%)	20	-0.404	-0.733	0.472	–	-1.416	0.678
Full sample	hike	Russell 2000 CAR (%)	20	-0.486	-0.716	0.483	–	-1.704	0.880
Full sample	hike	Financials CAR (%)	20	-0.917	-1.639	0.118	–	-1.958	0.209
Full sample	hike	Gold (%)	20	0.159	0.499	0.624	–	-0.452	0.764
Full sample	hike	Silver (%)	20	0.848	1.321	0.202	–	-0.427	2.071
Full sample	hike	Dollar (%)	20	0.089	0.388	0.702	–	-0.335	0.546
Full sample	hike	2Y yield (bp)	20	-3.150	-0.960	0.349	–	-9.650	2.950
Full sample	hike	5Y yield (bp)	20	-3.600	-1.099	0.285	–	-10.000	2.650
Full sample	hike	10Y yield (bp)	20	-2.900	-1.163	0.259	–	-7.550	1.900
Full sample	hike	30Y yield (bp)	20	-2.300	-1.429	0.169	–	-5.200	0.900
Full sample	hike	VIX (pts)	20	-0.189	-0.339	0.738	–	-1.245	0.845
Full sample	cut	S&P 500 CAR (%)	18	-0.201	-0.137	0.892	–	-3.122	2.470
Full sample	cut	Russell 2000 CAR (%)	18	-0.474	-0.253	0.803	–	-4.303	2.871
Full sample	cut	Financials CAR (%)	18	-0.010	-0.005	0.996	–	-4.238	3.458
Full sample	cut	Gold (%)	18	-0.608	-0.703	0.492	–	-2.313	0.982
Full sample	cut	Silver (%)	18	-0.726	-0.358	0.725	–	-4.901	2.773
Full sample	cut	Dollar (%)	18	-0.284	-0.697	0.495	–	-1.142	0.407
Full sample	cut	2Y yield (bp)	18	-1.889	-0.676	0.508	–	-7.222	3.278

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TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
Full sample	cut	5Y yield (bp)	18	-1.056	-0.295	0.772	–	-7.500	6.056
Full sample	cut	10Y yield (bp)	18	-1.000	-0.247	0.808	–	-8.722	6.667
Full sample	cut	30Y yield (bp)	18	-1.222	-0.358	0.725	–	-8.222	4.889
Full sample	cut	VIX (pts)	18	0.677	0.306	0.764	–	-3.175	5.254
Full sample	hold	S&P 500 CAR (%)	112	-0.096	-0.477	0.635	–	-0.496	0.278
Full sample	hold	Russell 2000 CAR (%)	112	-0.120	-0.449	0.654	–	-0.640	0.387
Full sample	hold	Financials CAR (%)	112	-0.152	-0.522	0.603	–	-0.734	0.410
Full sample	hold	Gold (%)	112	0.074	0.297	0.767	–	-0.419	0.560
Full sample	hold	Silver (%)	112	-0.313	-0.868	0.387	–	-1.024	0.390
Full sample	hold	Dollar (%)	112	-0.101	-1.080	0.282	–	-0.279	0.075
Full sample	**hold**	**2Y yield (bp)**	**112**	**-2.321**	**-2.644**	**0.009**	*******	**-4.098**	**-0.688**
Full sample	**hold**	**5Y yield (bp)**	**112**	**-3.062**	**-2.868**	**0.005**	*******	**-5.152**	**-1.027**
Full sample	**hold**	**10Y yield (bp)**	**112**	**-2.795**	**-2.478**	**0.015**	******	**-5.009**	**-0.607**
Full sample	hold	30Y yield (bp)	112	-1.938	-1.899	0.060	*	-3.973	0.062
Full sample	hold	VIX (pts)	112	0.051	0.162	0.872	–	-0.527	0.686
Full sample	all	S&P 500 CAR (%)	150	-0.150	-0.626	0.533	–	-0.633	0.314
Full sample	all	Russell 2000 CAR (%)	150	-0.211	-0.683	0.495	–	-0.832	0.382
Full sample	all	Financials CAR (%)	150	-0.237	-0.721	0.472	–	-0.919	0.381
Full sample	all	Gold (%)	150	0.003	0.015	0.988	–	-0.423	0.420
Full sample	all	Silver (%)	150	-0.207	-0.561	0.576	–	-0.960	0.488
Full sample	all	Dollar (%)	150	-0.098	-1.084	0.280	–	-0.279	0.071
Full sample	**all**	**2Y yield (bp)**	**150**	**-2.380**	**-2.806**	**0.006**	*******	**-4.047**	**-0.786**
Full sample	**all**	**5Y yield (bp)**	**150**	**-2.893**	**-2.900**	**0.004**	*******	**-4.827**	**-0.947**
Full sample	**all**	**10Y yield (bp)**	**150**	**-2.593**	**-2.544**	**0.012**	******	**-4.647**	**-0.600**
Full sample	**all**	**30Y yield (bp)**	**150**	**-1.900**	**-2.147**	**0.033**	******	**-3.653**	**-0.200**
Full sample	all	VIX (pts)	150	0.094	0.263	0.793	–	-0.587	0.822
GFC	cut	S&P 500 CAR (%)	7	1.134	0.376	0.720	–	-5.106	6.163
GFC	cut	Russell 2000 CAR (%)	7	1.136	0.316	0.763	–	-6.171	7.151
GFC	cut	Financials CAR (%)	7	1.565	0.333	0.751	–	-8.302	7.961
GFC	cut	Gold (%)	7	-0.621	-0.417	0.691	–	-3.268	2.059
GFC	cut	Silver (%)	7	0.934	0.329	0.753	–	-4.549	5.785
GFC	cut	Dollar (%)	7	-1.193	-1.343	0.228	–	-2.963	0.208
GFC	cut	2Y yield (bp)	7	-0.714	-0.129	0.902	–	-11.286	9.286
GFC	cut	5Y yield (bp)	7	0.286	0.038	0.971	–	-12.714	14.286
GFC	cut	10Y yield (bp)	7	-0.857	-0.091	0.930	–	-17.857	16.429
GFC	cut	30Y yield (bp)	7	-4.000	-0.528	0.617	–	-18.714	8.429
GFC	cut	VIX (pts)	7	-2.197	-0.687	0.518	–	-8.337	3.859
GFC	hold	S&P 500 CAR (%)	3	-2.649	-0.893	0.466	–	-7.791	2.492
GFC	hold	Russell 2000 CAR (%)	3	-2.651	-1.143	0.371	–	-6.461	1.542
GFC	hold	Financials CAR (%)	3	-3.707	-0.776	0.519	–	-12.549	3.841
GFC	hold	Gold (%)	3	3.385	0.809	0.504	–	-3.744	10.752
GFC	hold	Silver (%)	3	1.329	0.331	0.772	–	-5.945	7.922
GFC	hold	Dollar (%)	3	-0.429	-0.556	0.634	–	-1.288	1.111
GFC	hold	2Y yield (bp)	3	-28.000	-1.513	0.269	–	-59.000	5.000
GFC	hold	5Y yield (bp)	3	-19.667	-1.309	0.321	–	-45.000	7.000
GFC	hold	10Y yield (bp)	3	-12.000	-0.990	0.427	–	-33.000	9.000
GFC	hold	30Y yield (bp)	3	-7.333	-0.723	0.545	–	-24.000	11.000
GFC	hold	VIX (pts)	3	3.170	0.825	0.496	–	-2.340	10.560
GFC	all	S&P 500 CAR (%)	10	-0.001	-0.000	1.000	–	-4.403	3.989
GFC	all	Russell 2000 CAR (%)	10	-0.000	-0.000	1.000	–	-4.998	4.640
GFC	all	Financials CAR (%)	10	-0.017	-0.005	0.996	–	-7.268	5.713
GFC	all	Gold (%)	10	0.581	0.362	0.726	–	-2.277	3.673
GFC	all	Silver (%)	10	1.053	0.479	0.643	–	-3.197	5.119
GFC	all	Dollar (%)	10	-0.964	-1.484	0.172	–	-2.323	0.086
GFC	all	2Y yield (bp)	10	-8.900	-1.205	0.259	–	-24.000	3.600
GFC	all	5Y yield (bp)	10	-5.700	-0.798	0.445	–	-18.700	7.702

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TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
GFC	all	10Y yield (bp)	10	-4.200	-0.573	0.581	–	-17.802	9.300
GFC	all	30Y yield (bp)	10	-5.000	-0.859	0.413	–	-15.900	5.500
GFC	all	VIX (pts)	10	-0.587	-0.231	0.822	–	-5.481	3.997
ZIRP	hold	S&P 500 CAR (%)	55	-0.079	-0.259	0.797	–	-0.701	0.511
ZIRP	hold	Russell 2000 CAR (%)	55	0.040	0.100	0.920	–	-0.753	0.797
ZIRP	hold	Financials CAR (%)	55	0.005	0.012	0.991	–	-0.840	0.863
ZIRP	hold	Gold (%)	55	-0.294	-0.810	0.422	–	-1.005	0.412
ZIRP	hold	Silver (%)	55	-0.862	-1.606	0.114	–	-1.940	0.173
ZIRP	hold	Dollar (%)	55	0.009	0.060	0.953	–	-0.303	0.302
ZIRP	hold	2Y yield (bp)	55	-1.055	-1.426	0.160	–	-2.491	0.364
ZIRP	hold	5Y yield (bp)	55	-2.000	-1.269	0.210	–	-5.127	1.109
ZIRP	hold	10Y yield (bp)	55	-1.982	-1.061	0.293	–	-5.691	1.600
ZIRP	hold	30Y yield (bp)	55	-0.764	-0.458	0.649	–	-3.945	2.436
ZIRP	hold	VIX (pts)	55	0.119	0.287	0.775	–	-0.645	0.955
ZIRP	all	S&P 500 CAR (%)	55	-0.079	-0.259	0.797	–	-0.678	0.495
ZIRP	all	Russell 2000 CAR (%)	55	0.040	0.100	0.920	–	-0.730	0.787
ZIRP	all	Financials CAR (%)	55	0.005	0.012	0.991	–	-0.841	0.857
ZIRP	all	Gold (%)	55	-0.294	-0.810	0.422	–	-1.026	0.398
ZIRP	all	Silver (%)	55	-0.862	-1.606	0.114	–	-1.919	0.191
ZIRP	all	Dollar (%)	55	0.009	0.060	0.953	–	-0.296	0.307
ZIRP	all	2Y yield (bp)	55	-1.055	-1.426	0.160	–	-2.527	0.327
ZIRP	all	5Y yield (bp)	55	-2.000	-1.269	0.210	–	-5.037	1.018
ZIRP	all	10Y yield (bp)	55	-1.982	-1.061	0.293	–	-5.600	1.564
ZIRP	all	30Y yield (bp)	55	-0.764	-0.458	0.649	–	-4.073	2.364
ZIRP	all	VIX (pts)	55	0.119	0.287	0.775	–	-0.647	0.998
Normalization	hike	S&P 500 CAR (%)	9	-0.637	-1.399	0.199	–	-1.556	0.125
Normalization	hike	Russell 2000 CAR (%)	9	-0.667	-1.266	0.241	–	-1.656	0.294
Normalization	hike	Financials CAR (%)	9	-1.085	-1.750	0.118	–	-2.195	0.113
Normalization	hike	Gold (%)	9	-0.194	-0.356	0.731	–	-1.224	0.757
Normalization	hike	Silver (%)	9	-0.307	-0.322	0.756	–	-2.280	1.116
Normalization	hike	Dollar (%)	9	0.419	1.111	0.299	–	-0.263	1.120
Normalization	hike	2Y yield (bp)	9	1.556	0.803	0.445	–	-1.556	5.444
Normalization	hike	5Y yield (bp)	9	1.000	0.362	0.727	–	-3.444	6.444
Normalization	hike	10Y yield (bp)	9	-2.111	-1.101	0.303	–	-5.222	1.667
Normalization	**hike**	**30Y yield (bp)**	**9**	**-4.556**	**-4.549**	**0.002**	*******	**-6.333**	**-2.778**
Normalization	hike	VIX (pts)	9	0.552	0.684	0.513	–	-0.909	2.012
Normalization	hold	S&P 500 CAR (%)	20	0.220	0.867	0.397	–	-0.273	0.709
Normalization	hold	Russell 2000 CAR (%)	20	0.113	0.382	0.706	–	-0.464	0.661
Normalization	hold	Financials CAR (%)	20	-0.132	-0.482	0.635	–	-0.658	0.376
Normalization	hold	Gold (%)	20	0.647	1.927	0.069	*	0.003	1.289
Normalization	hold	Silver (%)	20	1.004	2.064	0.053	*	0.043	1.914
Normalization	hold	Dollar (%)	20	-0.325	-2.053	0.054	*	-0.631	-0.031
Normalization	hold	2Y yield (bp)	20	-2.100	-1.594	0.128	–	-4.750	0.300
Normalization	**hold**	**5Y yield (bp)**	**20**	**-3.150**	**-2.382**	**0.028**	******	**-5.650**	**-0.600**
Normalization	hold	10Y yield (bp)	20	-2.350	-2.085	0.051	*	-4.500	-0.200
Normalization	hold	30Y yield (bp)	20	-1.800	-1.667	0.112	–	-3.850	0.300
Normalization	hold	VIX (pts)	20	-0.468	-1.124	0.275	–	-1.221	0.362
Normalization	all	S&P 500 CAR (%)	29	-0.046	-0.197	0.845	–	-0.502	0.393
Normalization	all	Russell 2000 CAR (%)	29	-0.129	-0.485	0.631	–	-0.641	0.359
Normalization	all	Financials CAR (%)	29	-0.428	-1.550	0.132	–	-0.971	0.097
Normalization	all	Gold (%)	29	0.386	1.328	0.195	–	-0.166	0.946
Normalization	all	Silver (%)	29	0.598	1.321	0.197	–	-0.314	1.434
Normalization	all	Dollar (%)	29	-0.094	-0.558	0.581	–	-0.414	0.227
Normalization	all	2Y yield (bp)	29	-0.966	-0.864	0.395	–	-3.172	1.138
Normalization	all	5Y yield (bp)	29	-1.862	-1.460	0.156	–	-4.172	0.690
Normalization	**all**	**10Y yield (bp)**	**29**	**-2.276**	**-2.372**	**0.025**	******	**-4.070**	**-0.379**

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TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
Normalization	**all**	**30Y yield (bp)**	**29**	**-2.655**	**-3.190**	**0.003**	*******	**-4.172**	**-0.966**
Normalization	all	VIX (pts)	29	-0.151	-0.395	0.696	–	-0.873	0.608
Mid-cycle	cut	S&P 500 CAR (%)	3	-0.841	-0.975	0.432	–	-2.552	0.203
Mid-cycle	cut	Russell 2000 CAR (%)	3	-1.068	-3.957	0.058	*	-1.406	-0.534
Mid-cycle	cut	Financials CAR (%)	3	-1.285	-1.229	0.344	–	-3.368	-0.086
Mid-cycle	cut	Gold (%)	3	0.401	0.764	0.525	–	-0.313	1.426
Mid-cycle	cut	Silver (%)	3	-0.378	-0.486	0.675	–	-1.521	1.105
Mid-cycle	cut	Dollar (%)	3	-0.150	-0.592	0.614	–	-0.420	0.356
Mid-cycle	cut	2Y yield (bp)	3	-8.000	-2.000	0.184	–	-12.000	0.000
Mid-cycle	cut	5Y yield (bp)	3	-11.333	-2.713	0.113	–	-16.000	-3.000
Mid-cycle	cut	10Y yield (bp)	3	-12.333	-3.364	0.078	*	-16.000	-5.000
Mid-cycle	**cut**	**30Y yield (bp)**	**3**	**-13.667**	**-5.686**	**0.030**	******	**-17.000**	**-9.000**
Mid-cycle	cut	VIX (pts)	3	1.510	0.849	0.485	–	-0.620	5.040
Mid-cycle	**hold**	**S&P 500 CAR (%)**	**2**	**0.859**	**25.870**	**0.025**	******	**0.826**	**0.892**
Mid-cycle	hold	Russell 2000 CAR (%)	2	0.307	0.691	0.615	–	-0.137	0.752
Mid-cycle	hold	Financials CAR (%)	2	1.593	8.824	0.072	*	1.412	1.773
Mid-cycle	hold	Gold (%)	2	0.465	11.367	0.056	*	0.424	0.506
Mid-cycle	hold	Silver (%)	2	0.804	0.751	0.590	–	-0.267	1.876
Mid-cycle	hold	Dollar (%)	2	-0.169	-2.192	0.272	–	-0.246	-0.092
Mid-cycle	hold	2Y yield (bp)	2	0.000	0.000	1.000	–	-3.000	3.000
Mid-cycle	hold	5Y yield (bp)	2	0.500	0.091	0.942	–	-5.000	6.000
Mid-cycle	hold	10Y yield (bp)	2	1.500	0.273	0.830	–	-4.000	7.000
Mid-cycle	hold	30Y yield (bp)	2	2.000	0.667	0.626	–	-1.000	5.000
Mid-cycle	hold	VIX (pts)	2	-2.330	-5.683	0.111	–	-2.740	-1.920
Mid-cycle	all	S&P 500 CAR (%)	5	-0.161	-0.256	0.811	–	-1.387	0.728
Mid-cycle	all	Russell 2000 CAR (%)	5	-0.518	-1.315	0.259	–	-1.174	0.171
Mid-cycle	all	Financials CAR (%)	5	-0.134	-0.147	0.890	–	-1.818	1.266
Mid-cycle	all	Gold (%)	5	0.427	1.480	0.213	–	-0.018	0.975
Mid-cycle	all	Silver (%)	5	0.095	0.154	0.885	–	-0.949	1.203
Mid-cycle	all	Dollar (%)	5	-0.158	-1.118	0.326	–	-0.372	0.115
Mid-cycle	all	2Y yield (bp)	5	-4.800	-1.554	0.195	–	-10.200	0.600
Mid-cycle	all	5Y yield (bp)	5	-6.600	-1.617	0.181	–	-13.400	0.200
Mid-cycle	all	10Y yield (bp)	5	-6.800	-1.579	0.189	–	-13.800	0.200
Mid-cycle	all	30Y yield (bp)	5	-7.400	-1.776	0.150	–	-14.600	-0.200
Mid-cycle	all	VIX (pts)	5	-0.026	-0.019	0.986	–	-2.006	2.662
COVID	cut	S&P 500 CAR (%)	2	-4.384	-0.444	0.734	–	-14.266	5.498
COVID	cut	Russell 2000 CAR (%)	2	-8.343	-0.707	0.608	–	-20.137	3.450
COVID	cut	Financials CAR (%)	2	-5.005	-0.545	0.682	–	-14.180	4.171
COVID	cut	Gold (%)	2	-2.574	-0.349	0.786	–	-9.953	4.806
COVID	cut	Silver (%)	2	-11.111	-0.700	0.611	–	-26.989	4.767
COVID	cut	Dollar (%)	2	0.408	0.335	0.794	–	-0.808	1.624
COVID	cut	2Y yield (bp)	2	-16.500	-6.600	0.096	*	-19.000	-14.000
COVID	cut	5Y yield (bp)	2	-15.500	-10.333	0.061	*	-17.000	-14.000
COVID	cut	10Y yield (bp)	2	-10.000	-10.000	0.063	*	-11.000	-9.000
COVID	cut	30Y yield (bp)	2	3.000	3.000	0.205	–	2.000	4.000
COVID	cut	VIX (pts)	2	10.335	0.560	0.675	–	-8.120	28.790
COVID	hold	S&P 500 CAR (%)	14	-0.255	-0.376	0.713	–	-1.609	0.958
COVID	hold	Russell 2000 CAR (%)	14	-0.734	-0.675	0.511	–	-2.952	1.121
COVID	hold	Financials CAR (%)	14	-0.702	-0.628	0.541	–	-3.019	1.080
COVID	hold	Gold (%)	14	0.049	0.092	0.928	–	-0.999	1.001
COVID	hold	Silver (%)	14	0.253	0.238	0.815	–	-1.793	2.198
COVID	hold	Dollar (%)	14	-0.253	-1.284	0.222	–	-0.603	0.136
COVID	hold	2Y yield (bp)	14	-1.143	-1.125	0.281	–	-3.071	0.786
COVID	hold	5Y yield (bp)	14	-1.000	-0.565	0.582	–	-4.286	2.429
COVID	hold	10Y yield (bp)	14	-0.643	-0.295	0.773	–	-5.143	3.143
COVID	hold	30Y yield (bp)	14	-0.643	-0.283	0.781	–	-5.357	3.286

Continued on next page

TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
COVID	hold	VIX (pts)	14	0.373	0.237	0.816	–	-2.506	3.438
COVID	all	S&P 500 CAR (%)	16	-0.771	-0.680	0.507	–	-3.148	1.157
COVID	all	Russell 2000 CAR (%)	16	-1.685	-1.071	0.301	–	-4.995	0.903
COVID	all	Financials CAR (%)	16	-1.239	-0.928	0.368	–	-3.962	1.022
COVID	all	Gold (%)	16	-0.279	-0.328	0.747	–	-1.988	1.174
COVID	all	Silver (%)	16	-1.168	-0.592	0.563	–	-5.400	1.926
COVID	all	Dollar (%)	16	-0.170	-0.803	0.434	–	-0.554	0.249
COVID	all	2Y yield (bp)	16	-3.062	-1.916	0.075	*	-6.375	-0.188
COVID	all	5Y yield (bp)	16	-2.812	-1.419	0.176	–	-6.625	0.937
COVID	all	10Y yield (bp)	16	-1.812	-0.878	0.394	–	-5.938	1.814
COVID	all	30Y yield (bp)	16	-0.188	-0.094	0.927	–	-4.250	3.187
COVID	all	VIX (pts)	16	1.618	0.694	0.498	–	-2.354	6.497
Tightening	hike	S&P 500 CAR (%)	11	-0.213	-0.224	0.827	–	-1.879	1.681
Tightening	hike	Russell 2000 CAR (%)	11	-0.338	-0.285	0.782	–	-2.380	1.965
Tightening	hike	Financials CAR (%)	11	-0.780	-0.858	0.411	–	-2.444	0.978
Tightening	hike	Gold (%)	11	0.447	1.207	0.255	–	-0.184	1.192
Tightening	**hike**	**Silver (%)**	**11**	**1.792**	**2.259**	**0.047**	******	**0.454**	**3.392**
Tightening	hike	Dollar (%)	11	-0.180	-0.658	0.525	–	-0.667	0.343
Tightening	hike	2Y yield (bp)	11	-7.000	-1.248	0.240	–	-17.636	3.455
Tightening	hike	5Y yield (bp)	11	-7.364	-1.370	0.201	–	-17.273	2.820
Tightening	hike	10Y yield (bp)	11	-3.545	-0.814	0.435	–	-11.455	4.818
Tightening	hike	30Y yield (bp)	11	-0.455	-0.166	0.872	–	-5.455	4.727
Tightening	hike	VIX (pts)	11	-0.795	-1.063	0.313	–	-2.247	0.571
Tightening	hold	S&P 500 CAR (%)	2	-0.281	-0.147	0.907	–	-2.198	1.635
Tightening	hold	Russell 2000 CAR (%)	2	-2.282	-0.760	0.586	–	-5.284	0.720
Tightening	hold	Financials CAR (%)	2	0.578	0.576	0.667	–	-0.425	1.581
Tightening	hold	Gold (%)	2	-1.268	-0.908	0.531	–	-2.663	0.128
Tightening	hold	Silver (%)	2	-2.611	-1.177	0.448	–	-4.830	-0.393
Tightening	hold	Dollar (%)	2	-0.040	-0.028	0.982	–	-1.468	1.387
Tightening	hold	2Y yield (bp)	2	13.000	2.167	0.275	–	7.000	19.000
Tightening	hold	5Y yield (bp)	2	7.500	1.364	0.403	–	2.000	13.000
Tightening	hold	10Y yield (bp)	2	2.500	0.714	0.605	–	-1.000	6.000
Tightening	hold	30Y yield (bp)	2	-1.500	-3.000	0.205	–	-2.000	-1.000
Tightening	hold	VIX (pts)	2	0.040	0.073	0.954	–	-0.510	0.590
Tightening	all	S&P 500 CAR (%)	13	-0.223	-0.270	0.792	–	-1.679	1.377
Tightening	all	Russell 2000 CAR (%)	13	-0.637	-0.594	0.563	–	-2.517	1.514
Tightening	all	Financials CAR (%)	13	-0.571	-0.728	0.481	–	-2.008	0.925
Tightening	all	Gold (%)	13	0.183	0.468	0.648	–	-0.547	0.912
Tightening	all	Silver (%)	13	1.114	1.316	0.213	–	-0.436	2.743
Tightening	all	Dollar (%)	13	-0.158	-0.564	0.583	–	-0.676	0.381
Tightening	all	2Y yield (bp)	13	-3.923	-0.755	0.465	–	-14.000	5.538
Tightening	all	5Y yield (bp)	13	-5.077	-1.055	0.312	–	-14.077	3.771
Tightening	all	10Y yield (bp)	13	-2.615	-0.700	0.497	–	-9.538	4.463
Tightening	all	30Y yield (bp)	13	-0.615	-0.267	0.794	–	-4.846	3.769
Tightening	all	VIX (pts)	13	-0.666	-1.046	0.316	–	-1.882	0.501
Plateau	hold	S&P 500 CAR (%)	8	0.192	0.260	0.802	–	-1.161	1.527
Plateau	hold	Russell 2000 CAR (%)	8	0.608	0.518	0.620	–	-1.426	2.897
Plateau	hold	Financials CAR (%)	8	0.245	0.288	0.782	–	-1.260	1.859
Plateau	hold	Gold (%)	8	0.485	0.849	0.424	–	-0.576	1.529
Plateau	hold	Silver (%)	8	-0.012	-0.011	0.991	–	-1.738	1.981
Plateau	hold	Dollar (%)	8	-0.302	-1.108	0.305	–	-0.855	0.095
Plateau	**hold**	**2Y yield (bp)**	**8**	**-12.625**	**-2.961**	**0.021**	******	**-20.500**	**-5.125**
Plateau	**hold**	**5Y yield (bp)**	**8**	**-14.375**	**-2.785**	**0.027**	******	**-23.125**	**-4.247**
Plateau	**hold**	**10Y yield (bp)**	**8**	**-13.625**	**-2.570**	**0.037**	******	**-22.125**	**-3.250**
Plateau	hold	30Y yield (bp)	8	-11.875	-2.300	0.055	*	-20.625	-1.750
Plateau	hold	VIX (pts)	8	-0.079	-0.098	0.924	–	-1.545	1.348

Continued on next page

TABLE XI: Regime Summary One Sample T-Test

Regime	Decision	Metric	N	μ	T Stat.	P-value	Sig.	95% Low CI	95% High CI
Plateau	all	S&P 500 CAR (%)	8	0.192	0.260	0.802	–	-1.146	1.518
Plateau	all	Russell 2000 CAR (%)	8	0.608	0.518	0.620	–	-1.439	2.864
Plateau	all	Financials CAR (%)	8	0.245	0.288	0.782	–	-1.227	1.816
Plateau	all	Gold (%)	8	0.485	0.849	0.424	–	-0.515	1.513
Plateau	all	Silver (%)	8	-0.012	-0.011	0.991	–	-1.730	2.006
Plateau	all	Dollar (%)	8	-0.302	-1.108	0.305	–	-0.868	0.093
Plateau	**all**	**2Y yield (bp)**	**8**	**-12.625**	**-2.961**	**0.021**	******	**-20.750**	**-5.000**
Plateau	**all**	**5Y yield (bp)**	**8**	**-14.375**	**-2.785**	**0.027**	******	**-22.875**	**-4.375**
Plateau	**all**	**10Y yield (bp)**	**8**	**-13.625**	**-2.570**	**0.037**	******	**-22.250**	**-3.250**
Plateau	all	30Y yield (bp)	8	-11.875	-2.300	0.055	*	-20.500	-2.000
Plateau	all	VIX (pts)	8	-0.079	-0.098	0.924	–	-1.579	1.368
Easing	cut	S&P 500 CAR (%)	6	-0.045	-0.047	0.964	–	-1.775	1.617
Easing	cut	Russell 2000 CAR (%)	6	0.569	0.314	0.766	–	-2.973	3.512
Easing	cut	Financials CAR (%)	6	0.456	0.353	0.739	–	-1.870	2.692
Easing	cut	Gold (%)	6	-0.443	-0.641	0.550	–	-1.579	0.882
Easing	cut	Silver (%)	6	0.625	0.266	0.801	–	-3.206	5.030
Easing	cut	Dollar (%)	6	0.478	1.282	0.256	–	-0.182	1.124
Easing	cut	2Y yield (bp)	6	4.667	1.909	0.115	–	0.333	9.000
Easing	cut	5Y yield (bp)	6	7.333	2.554	0.051	*	2.333	12.500
Easing	**cut**	**10Y yield (bp)**	**6**	**7.500**	**2.611**	**0.048**	******	**2.500**	**12.667**
Easing	**cut**	**30Y yield (bp)**	**6**	**6.833**	**2.847**	**0.036**	******	**2.500**	**11.000**
Easing	cut	VIX (pts)	6	0.393	0.194	0.854	–	-2.858	4.402
Easing	hold	S&P 500 CAR (%)	8	-0.247	-0.848	0.425	–	-0.780	0.265
Easing	hold	Russell 2000 CAR (%)	8	-0.072	-0.186	0.858	–	-0.762	0.641
Easing	hold	Financials CAR (%)	8	-0.006	-0.020	0.985	–	-0.566	0.509
Easing	hold	Gold (%)	8	-0.205	-0.152	0.883	–	-2.801	1.993
Easing	hold	Silver (%)	8	-1.436	-0.749	0.478	–	-5.139	1.867
Easing	hold	Dollar (%)	8	0.291	1.069	0.321	–	-0.212	0.785
Easing	hold	2Y yield (bp)	8	1.875	0.710	0.501	–	-3.125	6.500
Easing	hold	5Y yield (bp)	8	0.250	0.109	0.916	–	-4.000	4.250
Easing	hold	10Y yield (bp)	8	-1.375	-0.819	0.440	–	-4.500	1.625
Easing	hold	30Y yield (bp)	8	-1.750	-1.024	0.340	–	-4.750	1.375
Easing	hold	VIX (pts)	8	-0.133	-0.286	0.783	–	-0.976	0.704
Easing	all	S&P 500 CAR (%)	14	-0.161	-0.380	0.710	–	-0.975	0.643
Easing	all	Russell 2000 CAR (%)	14	0.203	0.263	0.797	–	-1.323	1.614
Easing	all	Financials CAR (%)	14	0.192	0.347	0.734	–	-0.832	1.256
Easing	all	Gold (%)	14	-0.307	-0.384	0.707	–	-1.921	1.104
Easing	all	Silver (%)	14	-0.552	-0.379	0.711	–	-3.320	2.200
Easing	all	Dollar (%)	14	0.371	1.723	0.109	–	-0.038	0.775
Easing	all	2Y yield (bp)	14	3.071	1.696	0.114	–	-0.429	6.287
Easing	all	5Y yield (bp)	14	3.286	1.660	0.121	–	-0.429	7.000
Easing	all	10Y yield (bp)	14	2.429	1.261	0.230	–	-1.214	6.143
Easing	all	30Y yield (bp)	14	1.929	1.072	0.303	–	-1.429	5.357
Easing	all	VIX (pts)	14	0.093	0.107	0.916	–	-1.364	1.849

TABLE XII: Figure 9 Table Note for Rate Cut Data

Asset	N	μ	SE	95% CI_{lower}	95% CI_{Upper}	T Stat.	P-value	Sig.
S&P 500 (%)	18	-0.201	1.467	-3.077	2.674	-0.137	0.892	
Russell 2000 (%)	18	-0.474	1.869	-4.138	3.19	-0.253	0.803	
Financials (%)	18	-0.01	2.003	-3.935	3.916	-0.005	0.996	
Gold (%)	18	-0.608	0.865	-2.303	1.087	-0.703	0.492	
Silver (%)	18	-0.726	2.027	-4.699	3.247	-0.358	0.725	
Dollar (%)	18	-0.284	0.408	-1.084	0.515	-0.697	0.495	
2Y yield (bp)	18	-1.889	2.793	-7.364	3.586	-0.676	0.508	
5Y yield (bp)	18	-1.056	3.581	-8.073	5.962	-0.295	0.772	

Cross-Asset Reactions to FOMC Announcements

Asset	N	μ	SE	95% CI_{lower}	95% CI_{Upper}	T Stat.	P-value	Sig.
10Y yield (bp)	18	-1	4.044	-8.926	6.926	-0.247	0.808	
30Y yield (bp)	18	-1.222	3.413	-7.912	5.468	-0.358	0.725	
VIX (pts)	18	0.677	2.214	-3.664	5.017	0.306	0.764	

TABLE XIII: Figure 9 Table Note for Rate Hike Data

Asset	N	μ	SE	95% CI_{lower}	95% CI_{Upper}	T Stat.	P-value	Sig.
S&P 500 (%)	20	-0.404	0.551	-1.483	0.676	-0.733	0.472	
Russell 2000 (%)	20	-0.486	0.679	-1.818	0.845	-0.716	0.483	
Financials (%)	20	-0.917	0.559	-2.014	0.179	-1.639	0.118	
Gold (%)	20	0.159	0.318	-0.465	0.782	0.499	0.624	
Silver (%)	20	0.848	0.641	-0.41	2.105	1.321	0.202	
Dollar (%)	20	0.089	0.231	-0.363	0.541	0.388	0.702	
2Y yield (bp)	20	-3.15	3.282	-9.582	3.282	-0.96	0.349	
5Y yield (bp)	20	-3.6	3.275	-10.018	2.818	-1.099	0.285	
10Y yield (bp)	20	-2.9	2.494	-7.788	1.988	-1.163	0.259	
30Y yield (bp)	20	-2.3	1.609	-5.454	0.854	-1.429	0.169	
VIX (pts)	20	-0.189	0.556	-1.278	0.901	-0.339	0.738	

TABLE XIV: Figure 9 Table Note for Hold Cut Data

Asset	N	μ	SE	95% CI_{lower}	95% CI_{Upper}	T Stat.	P-value	Sig.
S&P 500 (%)	112	-0.096	0.201	-0.491	0.299	-0.477	0.635	
Russell 2000 (%)	112	-0.12	0.267	-0.644	0.404	-0.449	0.654	
Financials (%)	112	-0.152	0.292	-0.724	0.419	-0.522	0.603	
Gold (%)	112	0.074	0.248	-0.413	0.56	0.297	0.767	
Silver (%)	112	-0.313	0.36	-1.019	0.393	-0.868	0.387	
Dollar (%)	112	-0.101	0.094	-0.285	0.082	-1.08	0.282	
2Y yield (bp)	112	-2.321	0.878	-4.042	-0.601	-2.644	0.009	***
5Y yield (bp)	112	-3.063	1.068	-5.156	-0.969	-2.868	0.005	***
10Y yield (bp)	112	-2.795	1.128	-5.005	-0.584	-2.478	0.015	**
30Y yield (bp)	112	-1.938	1.02	-3.937	0.062	-1.899	0.06	*
VIX (pts)	112	0.051	0.313	-0.562	0.663	0.162	0.872	

[Note1] Note1. Note. $^{*}p < .10$. $^{**}p < .05$. $^{***}p < .01$.

[Note2] Note2. 'Full sample' refers to the entire 2008-2026 window pooled together (150 FOMC meetings).

[Note3] Note3. 'all' decision pools together the rate hike, cut, and hold policy into one sample.

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